

ACCEPTED MANUSCRIPT • OPEN ACCESS

## Runoff reduction from widespread implementation of rain gardens in cities through urban-resolving Earth system modeling

To cite this article before publication: Laura C Gray *et al* 2026 *Environ. Res.: Climate* in press <https://doi.org/10.1088/2752-5295/aead0f>

### Manuscript version: Accepted Manuscript

Accepted Manuscript is “the version of the article accepted for publication including all changes made as a result of the peer review process, and which may also include the addition to the article by IOP Publishing of a header, an article ID, a cover sheet and/or an ‘Accepted Manuscript’ watermark, but excluding any other editing, typesetting or other changes made by IOP Publishing and/or its licensors”

This Accepted Manuscript is © 2026 The Author(s). Published by IOP Publishing Ltd.



As the Version of Record of this article is going to be / has been published on a gold open access basis under a CC BY 4.0 licence, this Accepted Manuscript is available for reuse under a CC BY 4.0 licence immediately.

Everyone is permitted to use all or part of the original content in this article, provided that they adhere to all the terms of the licence <https://creativecommons.org/licenses/by/4.0>

Although reasonable endeavours have been taken to obtain all necessary permissions from third parties to include their copyrighted content within this article, their full citation and copyright line may not be present in this Accepted Manuscript version. Before using any content from this article, please refer to the Version of Record on IOPscience once published for full citation and copyright details, as permissions may be required. All third party content is fully copyright protected and is not published on a gold open access basis under a CC BY licence, unless that is specifically stated in the figure caption in the Version of Record.

View the [article online](#) for updates and enhancements.

1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
21  
22  
23  
24  
25  
26  
27  
28  
29  
30  
31  
32  
33  
34  
35  
36  
37  
38  
39  
40  
41  
42  
43  
44  
45  
46  
47  
48  
49  
50  
51  
52  
53  
54  
55  
56  
57  
58  
59  
60

Runoff reduction from widespread implementation  
of rain gardens in cities through urban-resolving  
Earth system modeling

Laura C. Gray<sup>1</sup>, Yifan Cheng<sup>1</sup>, Lei Zhao<sup>1,2,3\*</sup>,  
Ashlynn S. Stillwell<sup>1\*</sup>

<sup>1</sup>Civil and Environmental Engineering, The Grainger College of  
Engineering, University of Illinois Urbana-Champaign, Urbana, IL, USA.  
<sup>2</sup>Institute for Sustainability, Energy, and Environment, University of  
Illinois Urbana-Champaign, Urbana, IL, USA.  
<sup>3</sup>National Center for Supercomputing Applications, University of Illinois  
Urbana-Champaign, Urbana, IL, USA.

\*Corresponding author(s). E-mail(s): [leizhao@illinois.edu](mailto:leizhao@illinois.edu);  
[ashlynn@illinois.edu](mailto:ashlynn@illinois.edu);

Abstract

Green stormwater infrastructure (GSI) is widely regarded as an important nature-based solution to combat the combined effects of urbanization and climate change, yet its efficacy varies substantially across locations. GSI has not been evaluated at broader scales due to complexity in representing local hydrology in larger scale models. Using an urban-resolving Earth system model, we assess the suitability of rain gardens, a form of ground-based GSI, in runoff reduction in urban areas across the contiguous United States. We find strong influence of background climates on the runoff reduction potential across cities. Our results demonstrate effective runoff reduction in colder and wetter climates from widespread urban greening via installation of rain gardens compared with arid climates, possibly enhancing resilience and livability in certain cities. Most runoff reduction was due to enhanced urban evapotranspiration rather than increased infiltration. However, not all urban areas are optimal for GSI implementation, likely due to influence of background climate.

## Introduction

Climate change affects urban environments as built infrastructure continues to age and worldwide urban populations increase<sup>1</sup>. Urban flooding and runoff have become prominent issues for many cities and regions, arising from a combination of urban growth, development in floodplains, altered precipitation patterns, and increases in impervious surfaces<sup>2</sup>. In addition, cities face urban heat island effects and more temperature extremes<sup>3, 4, 5, 6</sup>. Urban infrastructures in particular are at risk from hydroclimatic extremes associated with climate change if they are stressed by age and demand, or are located near coastlines, storm tracks, rivers, and arid areas of vegetation<sup>7</sup>. Coupled together, these factors stress the urban landscape and create livability challenges.

Climate models consistently project that frequency, duration, and severity of hydroclimatic extremes will increase over this century under climate change<sup>8, 9</sup>. There is thus a pressing need to advance understanding of how climate adaptation strategies can mitigate the negative impacts from climate change coupled with urban development<sup>4, 10</sup>. Many modeling studies use mesoscale weather forecasting models, such as the Weather Research Forecasting Model (WRF)<sup>11</sup>, to investigate adaptation strategies such as green roofs, cool roofs, and urban reforestation for heat adaptation<sup>12, 13, 14, 15, 16</sup>, often with a focus on cooling benefits. However, mesoscale modeling has a high computational demand, with most studies focusing on events in single cities<sup>12</sup>. Utilizing global climate models (GCMs) to model urban adaptation strategies at a subgrid level reduces computational costs and promotes study of multiple urban areas on a global scale<sup>3, 17</sup>. Prior studies have used GCMs like the Community Earth System Model (CESM) to study the impacts of urban adaptation strategies such as green roofs, white roofs, reflective pavement, and street vegetation<sup>3, 18, 19, 20</sup>. However, the hydroclimatic effects of in-ground vegetated green stormwater infrastructure have rarely been assessed across scales, primarily due to a lack of representation of the complex hydrologic processes associated with those systems.

Green stormwater infrastructure (GSI) has been commonly proposed as one promising tactic to combat some of the effects of urbanization and climate change, such as greater urban flooding and decreased water quality, and increase resilience and livability<sup>21,22</sup>. GSI is unique in that it has potential to provide both hydrologic and thermodynamic benefits<sup>23,24,3</sup>. In-ground vegetated GSI systems provide benefits due to their mimicry of and connection to the natural hydrologic cycle<sup>24</sup>, including vegetation and hydrologic complexity, though these systems are not often well modeled across different locations. Ground-based GSI can capture rain where it falls, decreasing the amount of surface runoff, helping alleviate urban flooding issues, treating stormwater, and providing cooling through evapotranspiration, among other functions<sup>23,24,3</sup>. However, different types of GSI are not always suitable solutions due to prevailing weather patterns, soil type, and other factors<sup>25</sup>. Additionally, while GSI is a promising tactic, it also has environmental justice ramifications depending on how it is implemented<sup>26,27</sup>.

Typical studies determining GSI effectiveness and suitability are location-specific or completed on a small scale<sup>21,28</sup>. For this work, we consider local scale to refer to parcels to city scale, and regional scale to refer to scales larger than a single city. Local work is extensive and focuses on runoff reduction, evapotranspiration, infiltration, and by nature is context specific<sup>29,30,21</sup>. While upscaling work can be done to go from site-specific to city-level, it typically requires intimate knowledge of the sewer systems and other hydraulic infrastructure in place<sup>29</sup>. As such, local scale studies are better suited to evaluating site-specific performance than to generating broadly transferable findings. Regional-scale assessments, while less useful for individual site-specific performance prediction, do reveal general suitability patterns within different climate forcing contexts and allow for comparative analyses between locations. For this reason, it serves as a compelling complement to local scale work, providing an initial screening of potential implementation locations as well as identifying key processes and factors that should receive more attention in local studies. By identifying where

climatic conditions are broadly favorable or constraining, they can help municipalities prioritize locations for detailed watershed and neighborhood analyses and identify the processes that should receive particular attention during local design. For example, regional differences in rainfall intensity, wetness, and evaporative demand can inform local evaluations of storage capacity, drawdown time, soil properties, underdrain or overflow requirements, and vegetation selection. Therefore, regional-scale assessments narrow the design space, guide targeted data collection and modeling, and support more efficient allocation of planning resources before site-specific analyses are undertaken. There is thus a need for research completed at broader scales to understand regional impacts of widespread urban GSI implementation<sup>21, 25</sup>.

Additionally, despite its climate change adaptation potential, there remain barriers to widespread GSI implementation on political, economic, and technological levels<sup>31, 32, 24</sup>. Cities are generally concerned with the upfront and lifetime costs of GSI systems, as well as risk, and are apprehensive about conflicting with existing laws or city codes by installing GSI. From a technological standpoint, there is fear about the presumed lack of data on suitability of GSI systems across different environmental conditions. Modeling these systems on a broader scale allows for better understanding of potential regional climate effects, but also provides consistent results to use for comparison across studies and can promote city-to-city learning and knowledge transfers. Furthermore, modeling these systems is a useful approach to provide context on functional suitability for a given area or municipality, especially since GSI systems can be costly and difficult to maintain<sup>32, 33</sup>, making their justification to involved parties sometimes difficult.

Here, we present a novel, ground-based GSI parameterization in the form of a rain garden for CESM's Community Land Model Urban (CLMU) based on the parameters from a calibrated U.S. rain garden model<sup>34</sup> and rain garden soil mix recommendations

109 from the U.S. EPA<sup>35</sup>. We conducted a thorough validation of our GSI parameteriza-  
110 tion against published reports, with results demonstrating high credibility of the new  
111 scheme. A rain garden was chosen to better understand benefits provided by a more  
112 hydrologically complex, vegetated, and in-ground GSI system. Two separate simula-  
113 tions were run for a present period (2006–2015) at a 10-km resolution, with differences  
114 observed between the rain garden simulation and the control simulation. Our results  
115 illustrate urban runoff reduction opportunities for many urban areas in the United  
116 States due to widespread implementation of rain gardens.

117 **Methodology**

118 **Community Land Model Urban**

119 In this study, we utilized the Community Earth System Model (CESM)<sup>36</sup>. CESM is  
120 a fully-coupled, global-scale climate model hosted by the National Center for Atmo-  
121 spheric Research (NCAR) and researchers in the Earth system science community<sup>37</sup>.  
122 CESM contains seven components that can communicate through a coupler: the atmo-  
123 sphere, land, land ice, ocean, sea ice, river, and wave model<sup>37,36</sup>. Currently, CESM  
124 is one of the few GCMs with an urban land parameterization as part of its land  
125 model<sup>38,5</sup>, providing a key motivation for its use in this study. The Community Ter-  
126 restrial Systems Model (CTSM), the land component of CESM (formerly known as  
127 the Community Land Model (CLM)), focuses mainly on how terrestrial ecosystems  
128 and their various chemical, biological, and physical processes can affect and be affected  
129 by climate. Grid cells in CTSM are made up of five land units: vegetated, glacier,  
130 crop, lake, and urban. CTSM has been evaluated widely against both in situ and  
131 remote sensing observations around the globe<sup>39,40,41,42</sup>. The urban parameterization  
132 scheme embedded within CTSM (CLMU), simulates “local urban climate as an oper-  
133 ational component of a global climate model”<sup>43</sup>. CLMU models the energy, radiation,

134 water, and mass transfer between the lower atmosphere and land in urban environ-  
 135 ments<sup>43</sup>. More details on how CLMU and CTSM simulate subsurface-land-atmosphere  
 136 hydrologic processes can be found in Olseon et al.<sup>39</sup> and Lawrence et al.<sup>44</sup>, respectively.

137 Each grid cell in CTSM is composed of different percentages of each land unit. The  
 138 model computes outputs for each land unit and its allocated area, and for some output  
 139 variables, will calculate an area-weighted average over the whole grid cell area. CLMU  
 140 includes an embedded global urban surface dataset developed for global climate mod-  
 141 eling. This urban surface dataset provides information on present-day urban extent  
 142 and properties for every grid cell containing an “urban” subgrid land unit. Informa-  
 143 tion provided includes radiative, thermal, and morphological characteristics of urban  
 144 areas<sup>38</sup>.

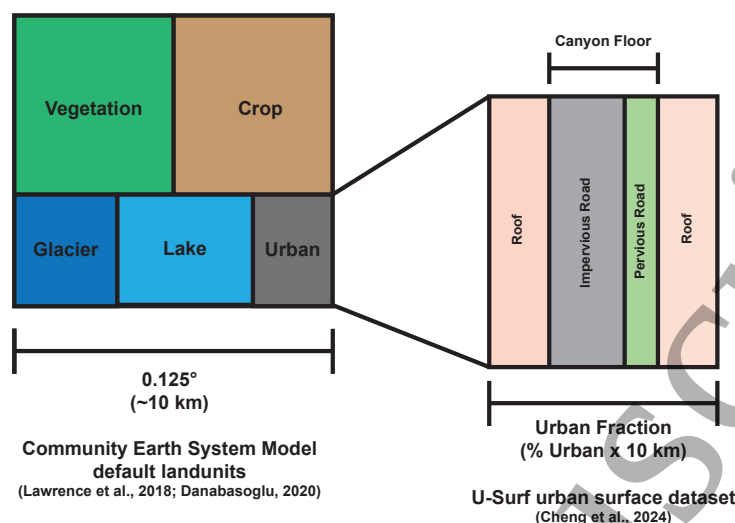
145 The urban scheme in CLMU is based on an “urban canyon” conceptual structure,  
 146 comprised of wall (sunlit and shaded), canyon floor (pervious and impervious), and  
 147 roof facets<sup>17</sup>. A schematic representation of CLMU and its role within the broader  
 148 gridcell is displayed in Figure 1. Importantly, Figure 1 illustrates that CLMU only  
 149 makes up only a portion of the overall gridcell. Adequate biophysical and hydrolog-  
 150 ical processes are captured over these facets<sup>17,43</sup>. For example, the pervious floor of  
 151 the canyon represents surfaces able to infiltrate water. Liquid water can pond on roofs  
 152 and canyon floors, supporting evaporation. Water in excess of the maximum ponding  
 153 depth or snowmelt water on both roofs and floors runs off. Pervious canyon floor evap-  
 154 oration is parameterized by a bulk scheme, where evaporation operates as a function of  
 155 the wetness of the entire soil column (to mimic vegetation roots) and water is removed  
 156 from each soil layer. A surface input dataset stipulating the radiative, thermal, mor-  
 157 phological, and hydrological characteristics of these facets for each grid cell in CTSM is  
 158 determined by satellite-derived data<sup>45</sup>. For the purposes of our simulations, values for  
 159 12 relevant urban surface property parameters from Cheng et al.’s 1-kilometer urban  
 160 surface dataset U-Surf<sup>46</sup> were implemented instead of the pre-existing surface dataset

161 driving CTSM to better represent urban areas. We utilized area weighting to aggregate  
162 U-Surf to the 0.125° resolution over CONUS and replaced the pre-existing values in  
163 the default Jackson et al.<sup>45</sup> configuration with U-Surf's updated values. Details of this  
164 new surface dataset and the updated variables can be found in Cheng et al.<sup>46</sup>. Addi-  
165 tionally, CLMU is forced using atmospheric model data (such as from the Community  
166 Atmosphere Model (CAM)) or from observed forcing (such as field observations or  
167 reanalysis data). To drive CLMU, air temperature and humidity, wind speed, precip-  
168 itation, and downward solar and longwave radiation from a reference height (e.g., the  
169 lowest level of the atmosphere model) are needed from the atmospheric forcing. CLMU  
170 then produces sensible heat, latent heat, momentum fluxes, and solar and longwave  
171 radiation budget, ground heat flux and heat storage, and anthropogenic heat flux that  
172 are area averaged with fluxes from nonurban landunits to provide gridcell-averaged  
173 fluxes to the atmosphere<sup>43</sup>.

174 Traditional studies on GSI often utilize local scale (i.e., neighborhood) models to  
175 examine hydrologic performance at a specific location. While such studies are useful  
176 from a hydrologic and runoff management perspective, they are limited in scope and  
177 are unable to provide information on broader suitability trends across locations and  
178 climates. While we do sacrifice some hydrologic precision by utilizing a GCM, we are  
179 still able to provide commentary on general rain garden suitability in varying climates  
180 and examine large-scale variations.

181 **Rain Garden Parameterization**

182 For this work, we use rain gardens as our representative GSI to understand bene-  
183 fits provided by a hydrologically complex, vegetated, and in-ground GSI system that  
184 was still able to be represented within the constraints of CLMU. To first create a  
185 functional parameterization within CLMU, we leveraged a local-scale calibrated rain  
186 garden model from William et al.<sup>34</sup>. Their calibrated parameters included thickness,



**Fig. 1:** Plan view schematic representation of landunit functionality within an example grid-cell with specific focus on the urban landunit. Each gridcell is made up of varying percentages of the five landunits: vegetation, crop, glacier, lake, and urban (shown on the left). The urban landunit component (right) includes a roof along each side of the canyon and does not overlap with the canyon floor, which includes the impervious road and the pervious road. Note that the pervious road typically makes up a smaller portion of the canyon floor than the impervious road. Included on the figure are also the main data sources for this analysis, with the land surface data coming from CESM's default settings, and urban inputs coming from the U-Surf dataset<sup>46</sup>.

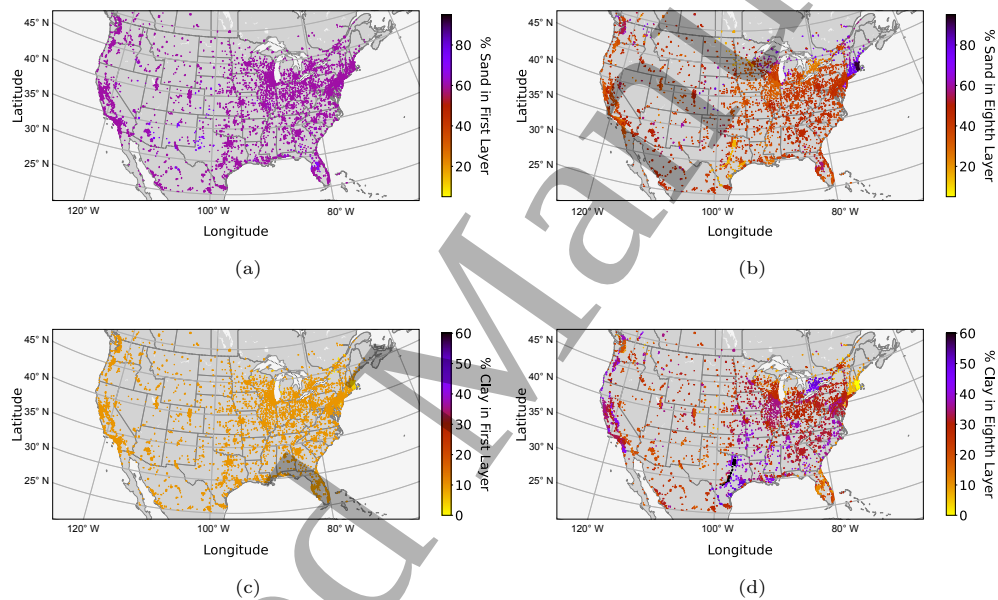
187 porosity, hydraulic conductivity, seepage rate, and void ratio<sup>34</sup>. In examining how the  
 188 surface and soil hydrology functioned in CLMU, specifically that of the pervious road,  
 189 we determined many relevant soil column calculations are related to porosity and con-  
 190 ductivity values<sup>17,44</sup>. However, the soil properties in CLMU are based on the more  
 191 simplistic soil data, only utilizing the percent sand and percent clay within the soil<sup>17</sup>.  
 192 CLMU's pervious road can function with a large number of soil layers in the hydrologic  
 193 column, but does not allow for explicit vegetation in the urban scheme at this time<sup>17</sup>.  
 194 Despite not having explicit vegetation representation, a bulk scheme accounting for  
 195 evaporation still exists and operates as a function of soil moisture<sup>17</sup>. Additionally,

CLMU does contain a rooting scheme<sup>17</sup>, despite a lack of explicit vegetation. While shading is an important consideration for other GSI strategies with taller vegetation, and is not currently represented in CLMU, typical rain gardens do not contain tall vegetation, allowing us to proceed with the parameterization.

We created and implemented a soil scheme representing rain garden installation within the model. In this research, the soil scheme for all pervious road areas in the urban landunit was altered to represent an optimistic scenario of widespread ground-based GSI implementation. This strategy of 100% conversion is commonly used in other climate adaptation modeling studies analyzing the suitability of green roofs, cool roofs, and street trees to bracket the upper bound of the potential effects of adaptation<sup>20,3,10,47</sup>. However, this work should not be viewed as a forecasting study, but instead more of an idealized, upper-bound sensitivity test and mechanistic study. We recognize this scenario is not practical, but can still provide contextual information on rain garden suitability across CONUS. For the urban soil columns, we implemented the porosity and saturated hydraulic conductivity values from the local-scale rain garden model<sup>34</sup> for the top seven layers of the soil column, corresponding to a depth of around 0.68 meters (approximately 27 inches), as a reasonable depth of engineered soil in a rain garden<sup>35,48,49,50</sup>. While these properties are derived from a single rain garden in Wisconsin, the soil in rain gardens is typically an engineered soil that can be designed to have ideal infiltration properties, such as those present in the rain garden in Wisconsin. Additionally, because the soil properties are based on percentages of sand and clay, we implemented 60% sand and 10% clay based on EPA and other common municipal guidelines for rain garden soil mixtures<sup>35,48,51,52,49,50</sup>. However, if the percent sand in one of the top seven layers of the urban soil column was already above 60%, we kept the remaining value in place, and if the percent clay was below 10%, we retained that value as well. The percent sand and percent clay in the remaining 13 layers remained their default values as prescribed in CESM, allowing us to capture

underlying soil trends. The default percent sand and percent clay values used by the model vary by location and soil layer and are derived from the International Geosphere-Biosphere Programme (IGBP) soil dataset (Global Soil Data Task 2000)<sup>44</sup>. CLMU assumes that urban soils do not have any organic matter present<sup>17</sup>, and as such any organic matter considerations are omitted from our parameterization. Figure 2 illustrates the soil properties (percent sand and percent clay) in representative soil layers within and outside of the rain garden parameterization across CONUS.

A schematic representation of these changes from the default hydrology of CLMU to the rain garden parameterization is shown in Figures 3(a) and 3(b), respectively. More information about how hydrology functions in CTSM and CLMU can be found in Oleson et al.<sup>17</sup> and Lawrence et al.<sup>44</sup>.



**Fig. 2:** Representation of the soil properties across CONUS in two representative layers of soil. Subfigures (a) and (c) represent the percent sand and percent clay respectively for the top layer of soil (located within the rain garden parameterization). Subfigures (b) and (d) represent the percent sand and percent clay respectively for the eighth layer of soil directly below the rain garden parameterization (reflecting native soil properties). Note that figures (a) and (c) reflect the 60% sand and 10% clay design mix, with exceptions for locations with percent sand greater than 60 and percent clay lesser than 10.

234 Additionally, to better simulate vegetation, we also altered the pervious road  
235 albedo for the rain garden simulation to have a value of 0.2. We assumed that rain  
236 garden vegetation could be approximated within the known albedos associated with  
237 “grass and pasture,” “green grass,” or “tall farm crops.” The 0.2 value falls within  
238 an appropriate range for all of these classifications<sup>53</sup>; we include a sensitivity analysis  
239 for albedo in the Supplemental Information with a summary in Table S1. We made  
240 this change in the surface dataset file used to run our rain garden simulation. While  
241 this parameterization was designed for rain gardens specifically, we note that the  
242 scheme could be easily adapted to other types of GSI such as planter boxes, infiltration  
243 trenches, or bioswales.

244 For further representation of the fraction and area of each of the 5466 CONUS  
245 gridcells that were altered due to the rain garden parameterization, see Supplemental  
246 Information Figures S4 and S5.

## 247 **Validation of Rain Garden Hydrology**

248 The CLMU itself has been evaluated extensively against remote sensing and in  
249 situ observations on urban temperature and surface energy balance across the  
250 globe<sup>28, 39, 40, 41, 54, 55, 56, 57, 58, 59</sup>. Previous work has validated CESM’s capability to  
251 robustly represent the hydrologic cycle and specifically total runoff by gridcell<sup>42</sup>.  
252 Because urban hydrologic data are scarce, urban runoff has not been directly validated.  
253 However, since the water balance and cycling are closely tracked in each landunit in  
254 CESM, including CLMU, Gray et al.’s evaluation, even in the absence of measured  
255 urban runoff data at a global scale, can provide confidence and indicative information  
256 in CESM’s urban hydrologic modeling<sup>42</sup>.

257 To ensure the robustness of our simulation results, we further validated our new  
258 rain garden parameterization against published reports. There is a robust and growing  
259 body of literature on rain garden pilot installations and experiments in the United

States, which aligns with our region of study. Runoff reduction in rain gardens can vary widely; e.g., 12–89%<sup>60,61,62</sup>. Given this variability, we focused on the water mass balance and hydrologic mechanisms within the pervious road through our rain garden parameterization:

$$\Delta S = \text{inflow} - \text{outflow} \quad (1)$$

where  $\Delta S$  represents the change in storage over the time period. Inflow into the rain garden can be defined as a combination of precipitation and overland flow or run-on, while outflow can be defined as a combination of infiltration, evapotranspiration, and runoff. Because overland flow is routed to the river scheme of CESM<sup>44</sup> and does not flow to the parameterized rain garden, the “inflow” term simplifies to only include precipitation. Here, we conducted a further CONUS land-only simulation with the rain garden parameterization for 1999–2018 to perform validations on key water balance components of rain gardens against present-day, local- and regional-scale, observational and/or modeling studies in the literature. The simulation is driven by the NLDAS-2 re-analysis dataset as the atmospheric forcing<sup>63</sup> to provide present-day results for validation with observations. It ran at the same spatial resolution (10 km) as our other simulations, and we analyzed daily average values to be consistent with literature.

We validated the rain garden parameterization with 14 case studies<sup>64,62,65,66,67,68,69,70,71,72,73,74,75,76</sup> using reported infiltration and evapotranspiration values as a percentage of incoming water. The reported rain garden percentages used for comparison to CLMU come from a variety of observational and modeling methods, are often reported in different units (e.g., mm/d, L, and percentages), and represent locations in the United States. Due to unit and scale

discrepancies, we used percentages to capture and compare underlying hydrologic trends. We calculated the model outputs of evapotranspiration and infiltration as a percentage of the total model precipitation in CLMU on the basis of information of the study areas to enable this comparison (see the footnotes of Supplemental Information Figure S11 for details).

Equations 2 and 3 were used to calculate percent infiltration and evapotranspiration:

$$\% \text{Infiltration} = \frac{\sum_{i=1}^n x_i}{\sum_{i=1}^n z_i} \times 100 \quad (2)$$

$$\% \text{Evapotranspiration} = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n z_i} \times 100 \quad (3)$$

where  $n$  represents the number of events for the time period,  $x$  represents infiltration volume,  $y$  represents evapotranspiration volume, and  $z$  represents the total precipitation for the rain garden. Infiltration and evapotranspiration are the primary mechanisms for reducing runoff in GSI, providing a key motivation for this study. All estimates were computed using grid cell(s) that contain the respective study regions. In some cases, the exact gridcell did not contain an urban component, so the next nearest gridcell with an active urban component was used for calculation. Additionally, all percentages from CESM were calculated with respect to total precipitation over the period, as the only hydrologic input to each gridcell and landunit is precipitation (CESM does not allow for run-on or inflow from other gridcells). However, the reported studies often included overland flow, run-on, or stormwater influent as part of the inflow into the rain garden, in addition to precipitation. These studies with influent included in the calculation are indicated in Supplemental Information Figure S11.

While this validation was focused on evapotranspiration and infiltration mechanisms as a result of incoming water (namely precipitation in CLMU), there are other components of the water balance that influence our results. In Supplemental Information Figure S11, some of our calculated evapotranspiration percentages exceed 100%. This result occurs during periods where there is more evapotranspiration than total precipitation, and is possible due to CLMU's soil moisture storage. Additionally, several of the studies in Supplemental Information Figure S11 used the terms "exfiltration" or "recharge" instead of "infiltration." For the sake of comparison, we assumed that these terms were comparable to CLMU's infiltration. Lastly, some studies included in Supplemental Information Figure S11 reported total inflow as well as precipitation alone; we report both for the sake of stronger comparison.

Almost half of the studies referenced in Supplemental Information Figure S11 come from Villanova, PA, USA, specifically Villanova University, and its instrumented campus rain gardens<sup>77,78</sup>. GSI functionality is an area of active research at an instrumented scale, with Villanova researchers completing a multitude of studies on these systems, hence the oversampling represented in our table. Additionally, since GSI systems tend to be designed primarily for their infiltration capabilities and flood reduction potential, performance data are usually reported in areas with localized flooding concerns. Thus, most available data on rain garden hydrology and water balances come from areas in the Midwest and Northeastern regions of the United States, which is reflected in our table. Studies on rain garden water balances, and not runoff reduction, are also limited as a field of ongoing study.

From Supplemental Information Figure S11, we show that CLMU modeled infiltration and evapotranspiration as a percentage of total precipitation are in reasonable agreement with the modeled and observed values based on comparison between observed and modeled infiltration and evapotranspiration from rain gardens across

seven United States cities. Almost all of the infiltration percentages are within reasonable range or in direct alignment with literature. Evapotranspiration percentages have more variability, but almost half of the entries still have good agreement. The existing differences could stem from the following reasons. (1) Inclusion of influent: many of the studies referenced in our table included influent, in addition to precipitation, as part of their inflow term. Studies that only used precipitation as their inflow into a rain garden tended to have stronger agreement with our CLMU calculations. Only using precipitation as an input is closer to our calculations. (2) Evapotranspiration calculations: there are a variety of methods to calculate and observe evapotranspiration, such as the Penman-Monteith (P-M) equation or the Hargreaves equation<sup>73,76</sup>, some of which are used in the selected case studies. CESM, and thus CLMU, parameterizes evapotranspiration utilizing the bulk transfer scheme and Monin-Obukhov Similarity Theory (MOST)<sup>44</sup>. While the underlying physics between the P-M equation and MOST, for example, are the same because the P-M equation is derived from MOST, they can still produce different estimates of evapotranspiration<sup>79</sup>. (3) Resolutions: while our simulation is striking in that it allows the global CESM/CLMU to run at  $0.125^{\circ}$  (10-km) fine resolution over CONUS, this resolution is still relatively coarse compared to the local-to-regional-scale studies, particularly studies of a single rain garden, which are typically not more than 30 square meters in size<sup>35</sup>. (4) Hydrologic Lag: for some of the studies, infiltration and evapotranspiration as a result of inflow are equal to 100%, particularly in controlled settings where no runoff is generated. This balance is not the case for our CLMU estimates, particularly as what is reported as infiltration can then become evapotranspiration. In CLMU's pervious road, there is a noticeable lack of hydrologic lag time (see Fig. 8 and 9). Due to this hydrologic lag, there is an increased likelihood that infiltration can become evapotranspiration in a short time period, particularly as evapotranspiration is parameterized in a bulk scheme that removes water as a function of the "wetness" of each soil layer.

Notably, a similar study from 2014 completed by Demuzere et. al implemented a bioinfiltration system with irrigation and drainage into CLMU<sup>28</sup>. This study is not included in our validation table, as it was completed in Australia, outside of the scope of our model. However, this study used an analogous methodology to the one adapted here, utilizing some altered parameter values (rain garden depth, porosity, and saturated conductivity) similar to our own. Additionally, they also focused on changes to soil properties, but implemented a separate soil scheme, instead of changing the percent sand and percent clay. Demuzere et. al completed this study and carried out a successful validation of soil moisture (95% confidence) for their location and time period<sup>28</sup>. Based on the similarities in studies, we assert that these results further illustrate the credibility of our study.

## Simulation Plan and Analysis

We ran two separate simulations: one control simulation, with the default CLMU in place, and one test simulation, with the rain garden parameterization. Both of these simulations were run at the finest resolution currently available in CESM of 10 km, or 0.125°, using the land-only setting. This resolution is a regional refined grid and only runs at the 10 km resolution over CONUS and at nominal 1 degree elsewhere. Land-only simulations require atmospheric climate forcing data to drive the land model; for these simulations, this atmospheric dataset came from Phase 2 of the North American Land Data Assimilation System (NLDAS-2)<sup>63</sup>. The 2006–2015 period was selected to represent the “current” climate and our surface dataset used Jackson et al.’s urban extent<sup>45</sup> with Cheng et al.’s<sup>46</sup> updated parameters. CESM calculates all variables at a time-step of 30 minutes, but only provides outputs at hourly, daily, and monthly intervals. For the sake of this analysis, we utilize the monthly outputs to examine longer temporal trends. A summary of the key parameters for the control simulation and the rain garden simulation is shown in Table 1.

**Table 1:** Parameter differences between the control and GSI simulation.

| Parameters                              | Rain Garden (GSI) Simulation              | Control (Default) Simulation                   |
|-----------------------------------------|-------------------------------------------|------------------------------------------------|
| Resolution                              | 10 km                                     | 10 km                                          |
| Time Period                             | 2006–2015                                 | 2006–2015                                      |
| Saturated hydraulic conductivity (Ksat) | 115 mm/hr<br>(for first 7 layers of soil) | Default value for soil properties of grid cell |
| Porosity                                | 0.44<br>(for first 7 layers of soil)      | Default value for soil properties of grid cell |
| % Sand                                  | 60%<br>(for first 7 layers of soil)       | Default value for soil properties of grid cell |
| % Clay                                  | 10%<br>(for first 7 layers of soil)       | Default value for soil properties of grid cell |
| Surface Albedo                          | 0.2                                       | Default for grid cell                          |
| Forcing Meteorology                     | NLDAS-2                                   | NLDAS-2                                        |

All simulations and data processing were conducted in NCAR’s High Performance Computing (HPC) cluster, Derecho<sup>80</sup>. Python was used to process simulation results. To complete city-specific analysis (e.g., for Chicago), we utilized shapefiles from the U.S. Census Bureau<sup>81</sup> and overlaid them with the ~10-km grid to investigate trends within specific city boundaries. Gridcells that overlapped with the shapefiles were flagged, and an average of the flagged gridcell values was computed to calculate the city mean. The cities of Chicago, New York, Seattle, and Atlanta were chosen for further analysis for several reasons. All four of these cities have documented issues with combined sewer overflows (CSOs), stemming from older combined sewer infrastructure. The U.S. Environmental Protection Agency (EPA) commonly recommends GSI as a solution to cities facing CSOs, hence why these four cities would be of note for rain garden suitability. Additionally, they represent four U.S. regions, and are all notably large urban areas (in terms of size and population). Because of their size, there are more urban gridcells for each of these cities, allowing for a more comprehensive analysis. Additionally, based on our results, we noted that most cities in arid areas saw little to no benefit in terms of runoff reduction, and did not think it was useful to perform further analysis on a representative arid city. The four cities chosen do represent a

blend of cold and temperate climates, with Chicago and parts of New York falling into the cold group, and Seattle, Atlanta, and other parts of New York in the temperate group. We specifically included Seattle due to the uniqueness of its results (increases in runoff) as opposed to other temperate cities.

For Köppen-Geiger climate analysis, we utilized gridded data for the present period provided by Beck et al.<sup>82</sup> and overlaid these data with our results for grouping and classification. One-way analysis of variance (ANOVA), Tukey pairwise testing, and one-tailed student's t-tests were then used to determine statistical significance of results.

## Results

### Runoff Reduction from Widespread Rain Gardens

The rain garden and control simulation results were compared using urban runoff (runoff generated only over the urban landunit) to evaluate suitability, since a key function of rain gardens is to increase infiltration and capture runoff from impervious areas<sup>35,76</sup>. Additionally, runoff represents a key hydrologic variable associated with human impacts<sup>42</sup>. For the purposes of this study, we focused on three runoff variables: urban total runoff,  $Q_t$ , urban surface runoff,  $Q_{srf}$ , and urban subsurface runoff,  $Q_{sub}$ .  $Q_{srf}$  represents the overland flow from saturation and infiltration excess mechanisms over the urban landunit, and  $Q_{sub}$  represents drainage due to saturated soil moisture conditions existing in the urban soil column<sup>17,44</sup>.  $Q_t$  represents the sum of  $Q_{srf}$  and  $Q_{sub}$ <sup>83</sup>.

In Figure 4, the average monthly difference, calculated over the 10-year period, between the control simulation and rain garden simulations' respective runoff values are shown for  $Q_{srf}$ ,  $Q_{sub}$ , and  $Q_t$ , respectively. When utilized, the subscript "c" refers to the control simulation, and "rg" refers to the rain garden simulation. Only gridcells with an urban landunit present are shown.

Figure 4(a) illustrates that many urban areas in the contiguous United States, on average, do not demonstrate much change in surface runoff as a result of the rain garden implementation. The mean surface runoff difference was -0.11 mm/month (where negative values indicate an average decrease in runoff) with a standard deviation of  $\pm 0.27$  mm/month and minimum and maximum values of -3.8 and 1.1 mm/month, respectively. These values span a small range and indicate minimal runoff reduction in most areas; however, some Midwestern and Northeastern states show a slightly larger negative difference between the rain garden and control simulations, indicating a small level of average runoff reduction. Around Massachusetts, a slight positive difference can be observed, indicating a possible average runoff increase when compared to the control simulation. This increase indicates that Massachusetts might not be a suitable location for rain garden implementation, or that it is not suited to this particular parameterization of GSI. The altered conditions created by the parameterization do not produce decreases in runoff, but rather increases, possibly due to the soil conditions created not being superior to what already existed in that location. Figure 4(b) shows a more pronounced difference in subsurface runoff than the difference in Figure 4(a)'s surface runoff in multiple urban areas, further illustrated by the mean subsurface runoff difference of -0.52 mm/month (a larger average decrease than for Figure 4(a); standard deviation  $\pm 1.1$  mm/month, minimum -10.7 mm/month, maximum 5.1 mm/month). These values span a larger range and indicate more subsurface runoff changes. Across the Midwest and Northeast, as well as in some urban areas bordering the Gulf of Mexico, there is a pronounced negative difference in subsurface runoff between the control and rain garden simulation, again indicating average runoff reduction. However, a few urban areas, specifically those located in the Pacific Northwest, show a positive difference, indicating a potential average runoff increase. Figure 4(c) illustrates the total urban runoff difference, with a mean difference of -0.63 mm/month (an average decrease in runoff; standard deviation  $\pm 1.1$  mm/month, minimum -10.0

mm/month, maximum 4.8 mm/month). Unsurprisingly, the strong signals from Figure 4(b) show up in Figure 4(c), with similar trends highlighted in Figure 4(c), particularly since the differences in subsurface runoff were larger than those in surface runoff. Overall, 84.5% of our urban gridcells see average decreases in total urban runoff. From a land area perspective,  $\sim 38,085 \text{ km}^2$  of the total  $\sim 48,486 \text{ km}^2$  urban area altered demonstrate a decrease in total runoff.

We then find percent total runoff reduction change between the rain garden simulation and control simulation utilizing Equations 4 and 5:

$$\text{Runoff Reduction} = (V_{in} - V_{out}) / (V_{in}) \quad (4)$$

$$\% \text{Change Runoff Reduction} = (RR_{rg} - RR_c) \times 100 \quad (5)$$

where  $V_{in}$  represents volume of inflow and  $V_{out}$  represents volume of outflow, adapted from Guo and Gao<sup>84</sup>. In our case, inflow simplifies to only precipitation, and outflow simplifies to the total urban runoff. We then calculate total urban runoff reduction for the rain garden simulation ( $RR_{rg}$ ) and for the control simulation ( $RR_c$ ), and subtract them from one another, shown in Equation 5. Using this methodology, we find a mean increase in total urban runoff reduction of  $0.8\% \pm 1.3\%$ , with a minimum value of  $-6.3\%$  (6.3% runoff increase) and maximum value of 10% increase in runoff reduction in response to the rain garden implementation.

We find strong climatic signals in the geographic patterns of runoff difference. We utilized the Köppen-Geiger climate classification<sup>82</sup> to investigate the impacts of cities' background climates on the simulated urban runoff changes in our results. All urban areas in the contiguous United States (CONUS) are broadly categorized into

three groups based their primary climate classification: cold, a combination of continental/cold and polar categories; temperate, a combination of temperate and tropical categories; and arid. We analyzed the mean total urban runoff difference over the 10-year period, with cities categorized into their respective three subgroups; Figure 5(a). Based on one-way analysis of variance (ANOVA) and Tukey pairwise testing, the runoff changes across the three climate groups are statistically significantly different from one another ( $p < 0.05$ ). Further examination using one-tailed student's t-tests confirms that cities in the "cold" subgroup, most of which are located in the Midwest and Northeast, show statistically significantly more runoff reduction than the temperate and arid groups ( $p < 0.05$  for both).

Previous studies have demonstrated that arid regions are more water-limited, meaning that water availability is the main limiting factor in ecological processes<sup>85,86</sup>, which explains why there is little runoff reduction in arid areas. Temperate and cold regions tend to be more humid, and are thus often energy-limited, with available energy (solar radiation) being the main constraint<sup>87,85</sup>. We theorize that despite both temperate and cold areas being energy-limited, cold regions see more runoff reduction due to their stronger seasonality. Cold regions in CONUS experience four strong seasons with wide temperature ranges, more so than temperate climates<sup>87</sup>. In these seasons, they fluctuate between available radiation and available water, unlike temperate and arid regions, which are almost always water-limited or energy-limited. This flexibility allows for more disturbances and thus responses to the existing surface hydroclimate.

In Figure 5(b), we further break down the runoff difference between the two simulations into exact Köppen-Geiger climate classifications. No polar CONUS gridcells have an urban landunit, so the four primary climate classes represented by U.S. cities are tropical ("A"), arid ("B"), temperate ("C"), and cold ("D"). For the exact breakdown of all of the climate zones listed in Figure 5(b), see Supplemental Information

Figure S10. Across all of the arid regions, there is little to no total urban runoff differences due to our rain garden implementation. Because arid regions are water-limited and have very little rain for most of the year, this result was not surprising. However, the tropical regions (comprising a small part of the tip of Florida), did see some slight runoff reduction as a result of our rain garden implementation. As shown in Figure 5(a), most of the cold regions see decreases in total urban runoff as a result of the implementation, with Dfa and Dfb, representing cold climates without dry seasons and hot and warm summers seeing the greatest decrease in total urban runoff. These regions comprise most of the American Midwest and Northeast, which is also consistent with our results in Figure 4. Most interestingly, climate zone Csb, representing temperate climates with dry, warm summers largely sees increases in total urban runoff. This climate zone comprises much of inner California as well as a large portion of the Pacific Northwest, which is again consistent with the results that we see in Figure 4. While other climate zones see some slight increases in total urban runoff, Csb is the only climate zone with a mean increase in total urban runoff. Other temperate zones see slight average decreases in total urban runoff.

Our results show that certain urban areas are more suitable candidates for installation of rain gardens based on our parameterization, and that reduction of runoff in both surface and subsurface runoff, on average, does not always occur in all urban locations. However, the results indicate that our parameterization does function as intended (capturing and reducing runoff) across many urban areas in the United States, and provides insight into areas potentially better suited for rain garden implementation. Our validation completed and shown in Figure S11 illustrates that the infiltration and evapotranspiration capabilities captured by our parameterization are consistent with independently observed and modeled rain garden results from literature. Additionally, approximately 78% of the urban land area saw runoff reduction as a result

of our implementation, even if it was minimal, further illustrating the effectiveness of the parameterization.

Figure 6 provides more detail about trends across four cities in four different U.S. regions (see Supplemental Information Figure S1 for city locations). Consistent with Figure 4, New York City and Chicago show a negative difference between scenarios (indicating runoff reduction), but Seattle notably shows a positive difference, particularly in the latter years of the simulation (indicating an increase in runoff). Atlanta shows almost no difference between simulations across the 10-year period, indicating our rain garden parameterization simulation provided minimal runoff reduction for that area. Interestingly, a seasonal pattern emerges regarding runoff differences between scenarios, where peak runoff reduction tends to occur in the winter months with occasional exceptions. However, Seattle shows a dramatically different pattern; through further investigation, we determined this difference was due to an increase in subsurface runoff, which might be due to greater infiltration from the rain garden system. However, this Seattle outcome is not observed for all cities, raising future research questions around effects of unique urban climate and form. Overall, the cities shown here are consistent with the findings that urban areas in the Midwest and Northeast typically show a greater runoff reduction benefit as a result of our rain garden parameterization in CLMU.

To understand the differences in water partitioning between the two simulations, we examined differences in other hydrologic variables. Figure 7(a) shows the difference in urban evapotranspiration ( $Q_{ET}$ ), and Figure 7(b) shows the difference in urban infiltration ( $Q_{infil}$ ). Note that the scale in Figure 7(b) is different than the other maps in this section and represents a smaller range of values. The mean evapotranspiration difference was 0.9 mm/month, illustrating an average increase in evapotranspiration, with a standard deviation of  $\pm 1.2$  mm/month and minimum and maximum values of -5.9 and 8.8 mm/month, respectively. Approximately 90% of our urban gridcells

see average increases in evapotranspiration, with  $\sim 43,959 \text{ km}^2$  showing increases in evapotranspiration of the total  $\sim 48,486 \text{ km}^2$  urban area altered. The mean infiltration difference was  $0.2 \text{ mm/month}$ , illustrating a small average increase in infiltration, with a standard deviation of  $\pm 0.3 \text{ mm/month}$  and minimum and maximum values of  $-1.2$  and  $4.0 \text{ mm/month}$ , respectively. Approximately 77% of our urban gridcells see increases in infiltration, with  $\sim 33,192 \text{ km}^2$  showing increases in infiltration of the total  $\sim 48,486 \text{ km}^2$  urban area altered. From Figure 7(a), we show that a large amount of incoming water (i.e., rainfall) partitions to evapotranspiration, as the areas with increased evapotranspiration (shown in blue, a positive difference) align closely with areas with less runoff (larger runoff reduction) in Figure 4(c), with similar magnitudes. Additionally, the areas in the Pacific Northwest with decreased evapotranspiration align closely with areas with more runoff in Figure 4(c). These results are consistent with monitored experimental rain gardens that show runoff reduction can vary between infiltration, evapotranspiration, and a combination of these mechanisms, depending on weather and infrastructure design<sup>75,71,76</sup>. However, Figure 7(b) (and supplementary Figures S2 and S3) illustrates that there is still some increased infiltration into the soil, also shown in blue, representing a positive difference. These areas of increased infiltration align with the areas of decreased surface runoff shown in Figure 4(a), which is consistent with infiltration-dominated GSI functionality. Because this parameterization uses implicit representation of vegetation, these results do not necessarily reflect survival or dormancy of plants in the winter months, nor do they differentiate between summer and winter processes outside of existing radiative effects. However, there are still seasonal trends in infiltration and evapotranspiration, illustrated further in the Supplemental Information Figures S6(c), S7(c), S8(c), and S9(c). Based on these select gridcells, evapotranspiration seasonality impact varies depending on location, while infiltration tends to increase in the winter months, likely linked to snowmelt. Infiltration of snowmelt into soil macropores and its successive refreezing can induce

frost heave. This induction creates new preferential flow pathways and expands existing pores in soil, allowing for deeper surface water infiltration and acceleration of soil thawing<sup>88</sup>.

## Runoff Dynamics During Storm Events

To understand the rain garden functionality in response to storm events, we selected two storm events for further study. Additional simulations over the storm periods were completed to output data on an hourly scale (not done for the overall 10-year simulation due to computational cost). In Figures 8 and 9, two storm events that occur over the city of Chicago, IL, are shown to illustrate urban runoff generation at an hourly scale. Respective precipitation is shown on the top horizontal axis of each set of figures. We selected a 14.5 mm (~0.5 in) spring storm event from April 19–20, 2011 (Figure 8), and a 35 mm (~1.4 in) winter storm event from January 29–30, 2013 (Figure 9), to compare differences in runoff generation during contrasting seasons. We selected Chicago as a reference city due to notable average runoff reduction in Figure 4 (see Supplemental Information Figure S1 for Chicago's location). During the spring storm, almost exclusively surface runoff is generated, with little to no subsurface runoff generated in both the control (Figure 8(a)) and rain garden simulations (Figure 8(b)). In Figure 8(c), there is a small negative difference between the two simulations, implying runoff reduction (likely due to enhanced infiltration, see supplementary Figures S2 and S3). During the winter storm, both surface and subsurface runoff are generated in the control and rain garden simulations (Figures 9(a) and 9(b), respectively), possibly in part due to snow or frozen soils. In Figure 9(c), there is a greater negative difference between the simulations almost uniformly across the 24-hour period, indicating runoff reduction. This reduction is also greater than that of the spring storm, which is consistent with the seasonality shown in our overall results.

## Discussion

For many municipalities and urban areas, GSI has been proposed as a promising climate adaptation strategy due to potential flood mitigation benefits. From a hydrologic perspective, GSI implementation can reduce surface water runoff, due largely to infiltration and evapotranspiration, and potentially alleviate urban flooding, if extensively implemented<sup>89,90</sup>. Vegetated GSI also can provide cooler “microclimate” areas due to evapotranspiration, affecting urban meteorology<sup>91</sup>. An increase in vegetative cover offers more possibilities for carbon sequestration, helps improve air quality, and also provides a habitat for biodiversity<sup>92</sup>. Because GSI offers a variety of benefits across sectors, it can be seen as a potentially appealing option for many municipalities with aging infrastructure. However, the results from this study illustrate that not all cities are suitable candidates for rain garden implementation from a hydrologic perspective. Based on this work, urban areas in the Midwest, Northeast, and those bordering the Gulf of Mexico would be more suitable areas to implement rain gardens for runoff reduction, whereas urban areas across most of the American West and Southeast show little to no runoff reduction across the 10-year study period. Some areas in the Pacific Northwest demonstrate increases in runoff in our simulations, likely primarily due to a decrease in evapotranspiration and corresponding increase in subsurface runoff.

A significant portion of the runoff reduction delivered by our parameterization was due to enhanced urban evapotranspiration rather than just increased infiltration. While most GSI is designed to reduce and capture runoff through infiltration, experimental studies have also demonstrated the importance of evapotranspiration in GSI water balances<sup>75,76</sup>. This combination of infiltration and evapotranspiration provides credence to our parameterization’s implicit vegetation representation and ability to mimic vegetative processes. Our results show that in a region without increased evapotranspiration (namely urban areas in Washington and Oregon) there was more subsurface runoff occurring, indicating that more water was present in the soil with

the implemented rain garden. This result is interesting, as it illustrates less total runoff reduction benefits provided in these areas, but that the parameterization was able to retain more water compared to the control. There was not necessarily much greater infiltration in these areas (see Fig. 7(b)), but due to the changes in soil hydrology, more water was captured in the soil. However, such a result does not indicate that rain gardens and GSI broadly are not a suitable adaptation strategy for areas such as the Pacific Northwest. The purpose of our work is to broadly understand suitability, and the results here motivate further studies completed on a local-urban watershed modeling scale with explicit considerations of underlying sewer systems that could better understand how GSI such as rain gardens would respond to actual implementation in a specific region, such as the Pacific Northwest. The Pacific Northwest has some of the most prevalent implementation of GSI in the United States, and the region has benefited from both water quantity and quality perspectives<sup>93</sup>. Most runoff reduction from our parameterization occurs during the winter, where evapotranspiration in the Pacific Northwest is likely lowest due to their cool, wet winters. While this area might not be evapotranspiring in the same manner as other locations in CONUS, water is still being retained in the soil and infiltration slightly increases due to the parameterization. Additionally, much of Seattle's sewer system is combined sewers, providing additional motivation to implement GSI to mitigate combined sewer overflows (CSOs). Thus, we assert that while the Pacific Northwest might not experience as many benefits due to evapotranspiration, it could still be a worthwhile investment from a water retention perspective and motivates further local-scale study. Additionally, while evapotranspiration played a major role, some regions of the Midwest and Northeast still saw increases in infiltration that particularly align with the decreases in surface runoff (Fig. 4(a) and 7(a)). These results demonstrate that our parameterization is thus representative of both infiltration and evapotranspiration mechanisms in vegetated GSI systems.

While we observed differences between the rain garden and control simulations in many urban areas, indicating the rain garden parameterization was functional, this change was smaller than expected. The local-scale reference rain garden was designed to infiltrate at least 1 in (25.4 mm) of runoff<sup>34</sup>, which was not observed in the results on a daily or monthly scale. Average runoff reduced for four of the cities across the 10-year period, with each city representing a U.S. geographic region, was less than 20 mm of urban runoff in any month over the 10-year period, indicating that the parameterization is limited, although functional. While 10 km (0.125°) is very fine resolution for a GCM, this spatial resolution is still coarse for representing local-scale hydrology. Each U.S. city is composed of multiple 10-km grid cells, likely contributing to this more “muted” runoff reduction signal.

We theorize that our results reflect artifacts of the model itself in addition to underlying climatic factors. Due to current limitations in CTSM and CLMU, the only hydrologic input into the land surface is precipitation, either from snow or rain. Runoff generated from other gridcells or subgrids within the gridcell (such as the impervious road within the urban landunit) is immediately routed to the river model, instead of flowing into another gridcell, landunit, or portion of the landunit. As such, our results only reflect direct precipitation interception, which further contextualizes this work as a suitability study of rain garden implementation rather than a full study on actual performance. Also because of this limitation in CTSM, our results should be considered as a conservative estimation of potential runoff reduction benefits. While the magnitude of benefits may be smaller due to this limitation, generally suitability should still be reflected in the results. Additionally, NLDAS-2, the dataset used for our atmospheric forcing, has known issues in underestimating high-intensity and convective summer precipitation<sup>94</sup>. As such, our runoff results, particularly in the summer, could potentially be biased.

The work presented in this study represents an optimistic scenario and brackets the upper-bound of potential impacts due to rain garden implementation. While not practical and a key limitation, this approach is still useful in providing information on rain garden suitability across CONUS. Ideally, a study could be completed with staged implementation scenarios, which could provide useful additional information for planning. However, such scenarios cannot be implemented in the current CTSM configuration by simply prescribing a lower conversion percentage. In our present implementation, the rain-garden parameterization replaces the existing pervious canyon-floor representation. Simulating partial conversion would require development of a new subgrid structure that allows conventional pervious canyon floor and engineered rain gardens to coexist within the same urban subgrid, together with the consistent aggregation of their water and energy fluxes and associated hydrologic routing. This would require substantial model development, testing, and validation and is therefore beyond the scope of the present study.

In our parameterization, we modified the surface albedo for the “pervious road” part of the landunit to a value of 0.2, which was higher than most default surface albedo values even with Cheng et al.’s<sup>46</sup> finer resolution surface dataset, U-Surf (see Figure S1). This change likely altered the partitioning of water in CLMU, thus increasing evapotranspiration in many areas. Although evapotranspiration accounts for most of the observed runoff reduction, increased infiltration was still observed in key areas in the Midwest and Northeast (shown in Figure 7(b)). The daily analysis in Chicago (as illustrated in Figure 8) showed that the current state of CLMU does not yet represent the lag time associated with runoff generation. Peak runoff directly correlated with peak precipitation, which is not reflective of rainfall-runoff relationships, even in urban areas<sup>95</sup>. Lag time is not present in either storm event shown in Figures 8 and 9, although runoff reduction did occur for both storms. We theorize this condition could be due to the simplicity of the current urban hydrology representation in CLMU, as

lag time is present in the vegetated landunits. CLMU and CESM are able to represent total runoff and the hydrologic cycle robustly<sup>42</sup>, but impervious surfaces in CLMU have no infiltration, only allow for ponding up to 1 millimeter, and send all surface water to runoff<sup>17</sup>. Since the urban component is made of impervious and pervious surfaces, it is not surprising to see little to no lag time with this representation. What little lag time that potentially exists in the pervious road is likely overshadowed by the lack of lag from the impervious road and roof, since the surface runoff generated over the urban landunit is a combination of runoff from the roof, pervious road, and impervious road (see Figure 1 for the schematic).

Additionally, while the overall Community Terrestrial Systems Model (CTSM), the land component of CESM, has CLMU embedded, vegetation is not explicitly represented within the urban landunit. This lack of vegetation in the urban landunit is one of the most significant limitations of our study, as our parameterization instead implicitly represents vegetation within the confines of the model's current version. Explicit vegetation representation does exist in the cropland and vegetated landunits of CTSM<sup>44</sup>, but has yet to be incorporated into the urban landunit. The urban landunit currently has a simplified surface and subsurface hydrology for the "pervious road," particularly regarding soil hydrology and assumed soil properties. A rooting scheme exists in CLMU's pervious road, but functions by distributing root mass equally through all hydrologically active layers and primarily serves as a mechanism to calculate evapotranspiration from each layer<sup>17</sup>. We altered the distribution of root mass in one version of our parameterization, but this altered rooting led to poor partitioning of water in lower soil layers, leading us to retain the default uniform distribution of roots. Lastly, because of this lack of explicit vegetation representation, we are unable to capture nuances in seasonality from vegetative processes, as the interactions between vegetation and background climate are simplified. We are not able to consider how available

1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
21  
22  
23  
24  
25  
26  
27  
28  
29  
30  
31  
32  
33  
34  
35  
36  
37  
38  
39  
40  
41  
42  
43  
44  
45  
46  
47  
48  
49  
50  
51  
52  
53  
54  
55  
56  
57  
58  
59  
60

739 water would impact plant function, dormancy, and survivability, a limitation of par-  
740 ticular concern since the results of this analysis point towards more runoff reduction  
741 potential during the winter months. As such, this study is likely not the precise tool  
742 to examine seasonality.

743 Most of the pervious road soil hydrology is based on empirical equations from a  
744 1980s study with three main properties: percent sand, percent clay, and percent organic  
745 matter present in a given layer<sup>44</sup>. These equations are further simplified for the urban  
746 landunit as CLMU assumes that urban soils have no organic matter present<sup>17</sup>, which  
747 is not representative of recent studies on urban soil<sup>96,97</sup>. This way of representing soil  
748 is somewhat simplistic, not accounting for complexities associated with different soil  
749 types and importantly, neglecting silt present in soils. The lack of organic material  
750 present in urban soils in CLMU also likely causes underestimation of infiltration and  
751 soil-water retention<sup>98,99</sup>. As a result, it is feasible that the exclusion of organic matter  
752 in soils potentially contributes to why our parameterization saw greater changes in  
753 evapotranspiration compared to infiltration. Additionally, the soil hydrology in the  
754 urban landunit uses the same soil properties as the natural vegetation in the gridcell  
755 and does not adjust to represent urban soils, outside of altering them to have no organic  
756 matter<sup>17</sup>. As such, the urban soils in CLMU do not reflect current soil conditions in  
757 cities, but do capture some location-specific effects. However, a similar study completed  
758 in 2014 implemented a bioinfiltration (or rain garden) parameterization into CLMU,  
759 with many of the authors' chosen parameters (such as rain garden depth, saturated  
760 hydraulic conductivity, and porosity) similar to ours. This work was completed and  
761 validated in only one location, but was still able to capture the nuances of a local rain  
762 garden<sup>28</sup>, reinforcing the applicability of our work.

763 Some nuances in our results are likely due to underlying contributors of location-  
764 specific weather patterns, climate, and soil conditions. Our simulations use historical  
765 weather data, Phase 2 of the North American Land Data Assimilation System

(NLDAS-2)<sup>63</sup>, to force the model, and the contiguous United States spans multiple climate zones according to the Köppen-Geiger climate classification system<sup>82</sup>. Many cities in continental climates show greater runoff reduction than those in other climate zones. This result could be because our rain garden parameterization was based on a calibrated rain garden model (built in the Environmental Protection Agency's Stormwater Management Model (EPA-SWMM)<sup>34</sup>) representing Wisconsin, which is also located in a continental climate. Additionally, we hypothesize that many continental/cold climates saw greater runoff reduction in the winter due to snowmelt. GSI hydrologic suitability in snow and cold conditions is an area of active research currently outside the scope of this study, but research from this area has shown that GSI is still active in the winter<sup>100,101</sup>, with one study finding that snow cover enabled soil temperatures to stay warm enough to continue infiltration<sup>101</sup>. We hypothesize that underlying soil conditions also influenced our results, similar to results from Sytsma et al.<sup>102</sup> showing calibration parameter transfer uncertainty in SWMM models with different soil, storm, and land cover forcings. Due to assumptions made in the rain garden design, almost all of the urban areas experienced major changes in percent sand and percent clay based on our parameterization for the top seven soil layers; however, the soil beyond the top seven layers likely played a role in deeper infiltration and seepage<sup>103</sup>. Prior work has found that urban soil properties converge to similar ranges across cities, likely as a result of management<sup>104</sup>, supporting our top seven soil layer assumption. Other work has indicated that properties such as urban soil hydraulic conductivity can vary across cities and locations<sup>105</sup>, however, which also could be contributing to rain garden suitability differences. Detailed investigation of urban soil properties is beyond the scope of this study.

790 **Conclusion**

791 Many cities understandably have concerns about implementing GSI systems into their  
792 municipal infrastructure. There is high uncertainty for many urban planners surround-  
793 ing GSI suitability for their specific location, particularly since GSI systems are much  
794 more variable in performance than their grey stormwater infrastructure counterparts.  
795 Additionally, for GSI systems to maintain functionality, they need to be properly main-  
796 tained and upkept (i.e., cleared of debris that can cause clogging, etc.)<sup>34</sup>. While our  
797 results provide information about the potential suitability of widespread rain gardens  
798 across many locations, the runoff reduction functionality of the systems does not con-  
799 sider the holistic socio-political context that could be useful for decision makers. Our  
800 work serves as a large-scale suitability assessment, identifying important processes and  
801 factors that should receive more attention in local scale studies, but lacks the ability  
802 to incorporate site-specific nuances. Other work has noted that GSI implementation  
803 can have environmental justice implications, particularly since access to green space  
804 is often inherently political and does have economic impacts as well<sup>26,27</sup>. Prior work  
805 advises that when making decisions about GSI implementation, urban planners and  
806 policymakers need to consider the impacts of not just the functionality of GSI systems,  
807 but how the location of GSI will impact the community and the different populations  
808 within the community<sup>26,27</sup>. Prior work also advises that community engagement with  
809 GSI planning and maintenance processes is vital, especially to ensure longevity and  
810 enhance resilience for the impacted communities<sup>106,107</sup>. Thus, while our results pro-  
811 vide insight into suitability of rain gardens across the contiguous United States, they  
812 only capture one aspect of GSI implementation and must be evaluated in context.

813 Despite these limitations, our rain garden parameterization successfully bridges  
814 microscale vegetation processes to a macroscale climate model. While some artifacts of  
815 the model are reflected in our results and some uncertainty exists in how we represent

rain gardens in CESM, we assert that the results presented still provide relevant information and commentary on applicability of rain gardens as an adaptation strategy. Our Köppen-Geiger climate classification analysis provides important information about the influence of background climate on potential rain garden suitability, and allows for stronger inter-region comparison. Our simulation results based on rain garden parameterization were generally consistent with observational and modeled studies of rain garden functionality (see Supplemental Information Figure S11). Parameterizing this kind of hydrologically-complex adaptation strategy into a GCM is an important step in model advancement to further provide a tool for cities to better prepare for climate change. Understanding how humans, and urban areas in general, affect climate is important for adaptation strategies. Through this work, we provide insight into how implementation of ground-based, vegetated GSI in the form of rain gardens functions in urban areas from a hydrologic perspective across the contiguous United States. While rain gardens are not a suitable adaptation strategy everywhere, they are still worth considering for runoff reduction benefits alone in the U.S. Midwest and Northeast, as well as in states bordering the Gulf of Mexico.

Through this work, we leveraged modeled data detailing potential hydrologic impacts of ground-based, vegetated GSI systems to advance knowledge on rain garden suitability for diverse areas and contexts. This research builds on previous studies completed on the mesoscale and local levels, and provides data on rain garden suitability trends on a larger scale. Such information could have important policy implications for U.S. municipal leaders weighing GSI investments, such as rain gardens, with environmental benefits for a specific location.

## 839 Acknowledgments

840 This work was supported in part under the provisions of section 104 of the Water  
 841 Resources Research Act annual base grants (104b) program made possible and dis-  
 842 tributed through the Illinois Water Resources Center and United States Geological  
 843 Survey. Additional support was provided by the Center for Infrastructure Resilience  
 844 in Cities as Livable Environments and the Support for Under-Represented Groups in  
 845 Engineering Fellowship from the Grainger College of Engineering at the University  
 846 of Illinois Urbana-Champaign. L.Z. acknowledges the support by the U.S. National  
 847 Aeronautics and Space Administration (NASA) through a joint Advanced Model-  
 848 ing Technology (AMT) and Land Cover Land Use Change (LCLUC) grant (Grant  
 849 80NSSC25K7322). We acknowledge the high-performance computing support from  
 850 Derecho (<https://www.cisl.ucar.edu/capabilities/derecho>) provided by NSF NCAR  
 851 Computational and Information Systems Laboratory, sponsored by the U.S. National  
 852 Science Foundation.

## 853 Author Contributions

854 L.C.G., L.Z., and A.S.S. conceptualized and developed the rain garden parameteri-  
 855 zation; L.C.G. implemented the parameterization and performed the data modeling  
 856 and analysis; L.Z. supervised the modeling; A.S.S. supervised the hydrologic analysis;  
 857 L.Z. and A.S.S. formulated the analysis, supervised the research, and obtained fund-  
 858 ing; Y.C. prepared the surface dataset used to drive all simulations; L.C.G. drafted  
 859 the manuscript, with discussions and contributions from L.Z. and A.S.S.

## References

- <sup>1</sup> United Nations. *World Urbanization Prospects 2018: Highlights* (2019). URL <https://www.un.org/development/desa/publications/2018-revision-of-world-urbanization-prospects.html>.
- <sup>2</sup> Ashley, R. M., Balmfort, D. J., Saul, A. J. & Blanskby, J. D. Flooding in the future - Predicting climate change, risks and responses in urban areas. *Water Science and Technology* **52**, 265–273 (2005).
- <sup>3</sup> Zhao, L., Lee, X. & Schultz, N. M. A wedge strategy for mitigation of urban warming in future climate scenarios. *Atmospheric Chemistry and Physics* **17**, 9067–9080 (2017). URL <https://acp.copernicus.org/articles/17/9067/2017/>.
- <sup>4</sup> Zhao, L. Urban growth and climate adaptation. *Nature Climate Change* **8**, 1034 (2018).
- <sup>5</sup> Zheng, Z., Zhao, L. & Oleson, K. W. Large model structural uncertainty in global projections of urban heat waves. *Nature Communications* **12** (2021).
- <sup>6</sup> Cutter, S. L. *et al.* Chapter 11: Urban Systems, Infrastructure, and Vulnerability. *Climate Change Impacts in the United States: The Third National Climate Assessment* (2014).
- <sup>7</sup> Wilbanks, T. J. & Fernandez, S. J. *Climate Change and Infrastructure, Urban Systems, and Vulnerabilities: Technical Report for the U.S. Department of Energy in Support of the National Climate Assessment* (2014).
- <sup>8</sup> Horton, R. M., Mankin, J. S., Lesk, C., Coffel, E. & Raymond, C. A Review of Recent Advances in Research on Extreme Heat Events. *Current Climate Change Reports* **2**, 242–259 (2016).

- <sup>9</sup> IPCC. *Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* (2014).
- <sup>10</sup> Krayenhoff, E. S., Moustauoui, M., Broadbent, A. M., Gupta, V. & Georgescu, M. Diurnal interaction between urban expansion, climate change and adaptation in US cities. *Nature Climate Change* **8**, 1097–1103 (2018). URL [wed](#).
- <sup>11</sup> Skamarock, W. C. *et al.* A Description of the Advanced Research WRF Version 3 (2008).
- <sup>12</sup> Li, D., Bou-Zeid, E. & Oppenheimer, M. The effectiveness of cool and green roofs as urban heat island mitigation strategies. *Environmental Research Letters* **9**, 055002 (2014). URL <https://dx.doi.org/10.1088/1748-9326/9/5/055002>.
- <sup>13</sup> Huber, M. Can a roof's material cool the outside air and lower energy demand? | Argonne National Laboratory (2023). URL <https://www.anl.gov/article/can-a-roofs-material-cool-the-outside-air-and-lower-energy-demand>.
- <sup>14</sup> Georgescu, M., Morefield, P. E., Bierwagen, B. G. & Weaver, C. P. Urban adaptation can roll back warming of emerging megapolitan regions. *Proceedings of the National Academy of Sciences* **111**, 2909–2914 (2014). URL <https://www.pnas.org/doi/10.1073/pnas.1322280111>.
- <sup>15</sup> Salamanca, F., Georgescu, M., Mahalov, A., Moustauoui, M. & Martilli, A. Citywide Impacts of Cool Roof and Rooftop Solar Photovoltaic Deployment on Near-Surface Air Temperature and Cooling Energy Demand. *Boundary-Layer Meteorology* **161**, 203–221 (2016).
- <sup>16</sup> Stone, B. J., Vargo, J., Liu, P., Hu, Y. & Russell, A. Climate Change Adaptation Through Urban Heat Management in Atlanta, Georgia. *Environmental Science &*

- 907 *Technology* **47**, 7780–7786 (2013). URL <https://doi.org/10.1021/es304352e>.
- 908 <sup>17</sup> Oleson, K. W., Bonan, G. B., Feddema, J. J., Vertenstein, M. & Kluzek,  
909 E. Technical Description of an Urban Parameterization for the Com-  
910 munity Land Model (CLMU). *NCAR/TN-480+STR NCAR Technical*  
911 *Note* (2010). URL [https://www.researchgate.net/profile/Keith-Oleson/](https://www.researchgate.net/profile/Keith-Oleson/publication/261062625_Technical_Description_of_an_Urban_Parameterization_for_the_Community_Land_Model_CLMU/links/02e7e5331acf6834e7000000/Technical-Description-of-an-Urban-Parameterization-for-the-Communit)  
912 [publication/261062625\\_Technical\\_Description\\_of\\_an\\_Urban\\_Parameterization\\_](https://www.researchgate.net/profile/Keith-Oleson/publication/261062625_Technical_Description_of_an_Urban_Parameterization_for_the_Community_Land_Model_CLMU/links/02e7e5331acf6834e7000000/Technical-Description-of-an-Urban-Parameterization-for-the-Communit)  
913 [for\\_the\\_Community\\_Land\\_Model\\_CLMU/links/02e7e5331acf6834e7000000/](https://www.researchgate.net/profile/Keith-Oleson/publication/261062625_Technical_Description_of_an_Urban_Parameterization_for_the_Community_Land_Model_CLMU/links/02e7e5331acf6834e7000000/Technical-Description-of-an-Urban-Parameterization-for-the-Communit)  
914 [Technical-Description-of-an-Urban-Parameterization-for-the-Communit](https://www.researchgate.net/profile/Keith-Oleson/publication/261062625_Technical_Description_of_an_Urban_Parameterization_for_the_Community_Land_Model_CLMU/links/02e7e5331acf6834e7000000/Technical-Description-of-an-Urban-Parameterization-for-the-Communit).
- 915 <sup>18</sup> Anderson, G. B., Oleson, K. W., Jones, B. & Peng, R. D. Projected trends  
916 in high-mortality heatwaves under different scenarios of climate, population, and  
917 adaptation in 82 US communities. *Climatic Change* **146**, 455–470 (2018). URL  
918 <https://doi.org/10.1007/s10584-016-1779-x>.
- 919 <sup>19</sup> Oleson, K. W., Bonan, G. B. & Feddema, J. Effects of white roofs on urban  
920 temperature in a global climate model. *Geophysical Research Letters* **37** (2010).  
921 URL <https://onlinelibrary.wiley.com/doi/abs/10.1029/2009GL042194>.
- 922 <sup>20</sup> Zhang, J., Zhang, K., Liu, J. & Ban-Weiss, G. Revisiting the climate impacts  
923 of cool roofs around the globe using an Earth system model. *Environmental*  
924 *Research Letters* **11**, 084014 (2016). URL [https://dx.doi.org/10.1088/1748-9326/](https://dx.doi.org/10.1088/1748-9326/11/8/084014)  
925 [11/8/084014](https://dx.doi.org/10.1088/1748-9326/11/8/084014).
- 926 <sup>21</sup> Prudencio, L. & Null, S. E. Stormwater management and ecosystem services: A  
927 review. *Environmental Research Letters* **13**, 033002 (2018). URL [https://dx.doi.](https://dx.doi.org/10.1088/1748-9326/aaa81a)  
928 [org/10.1088/1748-9326/aaa81a](https://dx.doi.org/10.1088/1748-9326/aaa81a).

- 22 Suppakittpaisarn, P., Jiang, X. & Sullivan, W. C. Green Infrastructure, Green  
Stormwater Infrastructure, and Human Health: A Review. *Current Landscape Ecology Reports* **2**, 96–110 (2017). URL <https://doi.org/10.1007/s40823-017-0028-y>.
- 23 Denchak, M. Green Infrastructure: How to Manage Water in  
a Sustainable Way (2019). URL [https://www.nrdc.org/stories/  
green-infrastructure-how-manage-water-sustainable-way](https://www.nrdc.org/stories/green-infrastructure-how-manage-water-sustainable-way).
- 24 US EPA. What is Green Infrastructure? (2023). URL [https://www.epa.gov/  
green-infrastructure/what-green-infrastructure](https://www.epa.gov/green-infrastructure/what-green-infrastructure).
- 25 Copeland, C. Green Infrastructure and Issues in Managing Urban Stormwater.  
CRS Report R43131, Congressional Research Service (2016). URL [https://sgp.fas.  
org/crs/misc/R43131.pdf](https://sgp.fas.org/crs/misc/R43131.pdf).
- 26 Hoover, F.-A., Meerow, S., Grabowski, Z. J. & McPhearson, T. Environmental  
justice implications of siting criteria in urban green infrastructure planning. *Journal  
of Environmental Policy & Planning* (2021). URL [https://www.tandfonline.com/  
doi/abs/10.1080/1523908X.2021.1945916](https://www.tandfonline.com/doi/abs/10.1080/1523908X.2021.1945916).
- 27 Meerow, S. & Newell, J. P. Urban resilience for whom, what, when, where, and  
why? *Urban Geography* **40**, 309–329 (2019). URL [https://www.tandfonline.com/  
doi/full/10.1080/02723638.2016.1206395](https://www.tandfonline.com/doi/full/10.1080/02723638.2016.1206395).
- 28 Demuzere, M. *et al.* The implementation of biofiltration systems, rainwater  
tanks and urban irrigation in a single-layer urban canopy model. *Urban Cli-  
mate* **10**, 148–170 (2014). URL [https://www.sciencedirect.com/science/article/  
pii/S2212095514000844](https://www.sciencedirect.com/science/article/pii/S2212095514000844).
- 29 Zhang, K. & Chui, T. F. M. Linking hydrological and bioecological benefits of  
green infrastructures across spatial scales – A literature review. *Science of The*

- 953 *Total Environment* **646**, 1219–1231 (2019). URL [https://linkinghub.elsevier.com/](https://linkinghub.elsevier.com/retrieve/pii/S004896971832850X)  
954 [retrieve/pii/S004896971832850X](https://linkinghub.elsevier.com/retrieve/pii/S004896971832850X).
- 955 <sup>30</sup> Lisenbee, W. A., Hathaway, J. M., Burns, M. J. & Fletcher, T. D. Model-  
956 ing bioretention stormwater systems: Current models and future research needs.  
957 *Environmental Modelling & Software* **144**, 105146 (2021). URL [https://www.](https://www.sciencedirect.com/science/article/pii/S1364815221001894)  
958 [sciencedirect.com/science/article/pii/S1364815221001894](https://www.sciencedirect.com/science/article/pii/S1364815221001894).
- 959 <sup>31</sup> Congressional Research Service. Green Infrastructure and Issues in Managing  
960 Urban Stormwater. Tech. Rep., Congressional Research Service (2016).
- 961 <sup>32</sup> Nemo, L. As Cities' Interest in Green Infrastructure Grows, So Does the Need to  
962 Develop Strategies and Resources to Maintain It. *Ensia* (2019). URL [https://ensia.](https://ensia.com/features/green-infrastructure-maintenance-flooding-pollution-groundwater/)  
963 [com/features/green-infrastructure-maintenance-flooding-pollution-groundwater/](https://ensia.com/features/green-infrastructure-maintenance-flooding-pollution-groundwater/).
- 964 <sup>33</sup> Water Environment Federation. The Real Cost of Green Infrastructure (2015).  
965 URL <https://stormwater.wef.org/2015/12/real-cost-green-infrastructure/>.
- 966 <sup>34</sup> William, R., Gardoni, P. & Stillwell, A. S. Reliability-Based Approach to Inves-  
967 tigating Long-Term Clogging in Green Stormwater Infrastructure. *Journal of*  
968 *Sustainable Water in the Built Environment* **5** (2019).
- 969 <sup>35</sup> Georgia, C. W. C. Rain Gardens for Home Landscapes. URL [https://cfpub.epa.](https://cfpub.epa.gov/npstbx/files/cwc_raingardenbrochure.pdf)  
970 [gov/npstbx/files/cwc\\_raingardenbrochure.pdf](https://cfpub.epa.gov/npstbx/files/cwc_raingardenbrochure.pdf).
- 971 <sup>36</sup> Danabasoglu, G. *et al.* The Community Earth System Model Version 2 (CESM2).  
972 *Journal of Advances in Modeling Earth Systems* **12** (2020).
- 973 <sup>37</sup> Hurrell, J. W. *et al.* The Community Earth System Model: A Framework for  
974 Collaborative Research. *Bulletin of the American Meteorological Society* **94**, 1339–  
975 1360 (2013).

- 38 Zhao, L. *et al.* Global multi-model projections of local urban climates. *Nature*  
39 *Climate Change* **11** (2021).
- 40 Oleson, K. W., Bonan, B., Feddema, J., Vertenstein, M. & Grimmond, C. S. An  
41 urban parameterization for a global climate model. Part 1: Formulation and evaluation for two cities. *Journal of Applied Meteorology and Climatology* **47**, 1038–1060  
42 (2008).
- 43 Zhao, L., Lee, X., Smith, R. B. & Oleson, K. Strong contributions of local  
44 background climate to urban heat islands. *Nature* **511**, 216–219 (2014).
- 45 Demuzere, M. *et al.* Impact of urban canopy models and external parameters on  
46 the modelled urban energy balance in a tropical city. *Quarterly Journal of the*  
47 *Royal Meteorological Society* **143**, 1581–1596 (2017).
- 48 Gray, L. C., Zhao, L. & Stillwell, A. S. Impacts of climate change on global  
49 total and urban runoff. *Journal of Hydrology* **620**, 129352 (2023). URL <https://www.sciencedirect.com/science/article/pii/S0022169423002949>.
- 50 Oleson, K. W. & Feddema, J. Parameterization and Surface Data Improvements  
51 and New Capabilities for the Community Land Model Urban (CLMU). *Journal of Advances in Modeling Earth Systems* **12**, e2018MS001586 (2020). URL <https://onlinelibrary.wiley.com/doi/abs/10.1029/2018MS001586>. E2018MS001586  
52 2018MS001586.
- 53 Lawrence, D. *et al.* Technical Description of version 5.0 of the Community Land  
54 Model (CLM). *NCAR/TN-478+STR NCAR Technical Note* (2018).
- 55 Jackson, T. L., Feddema, J. J., Oleson, K. W., Bonan, G. B. & Bauer, J. T.  
56 Parameterization of urban characteristics for global climate modeling. *Annals of*  
57 *the Association of American Geographers* **100** (2010).

- 1000 <sup>46</sup> Cheng, Y. *et al.* U-Surf: A global 1 km spatially continuous urban surface property  
1001 dataset for kilometer-scale urban-resolving Earth system modeling. *Earth System*  
1002 *Science Data* **17**, 2147–2174 (2025). URL [https://essd.copernicus.org/articles/17/](https://essd.copernicus.org/articles/17/2147/2025/)  
1003 [2147/2025/](https://essd.copernicus.org/articles/17/2147/2025/).
- 1004 <sup>47</sup> Jiang, T. *et al.* Prioritizing urban heat adaptation infrastructure based on multi-  
1005 ple outcomes: Comfort, health, and energy. *Proceedings of the National Academy*  
1006 *of Sciences* **122**, e2411144122 (2025). URL [https://www.pnas-org.proxy2.library.](https://www.pnas-org.proxy2.library.illinois.edu/doi/10.1073/pnas.2411144122)  
1007 [illinois.edu/doi/10.1073/pnas.2411144122](https://www.pnas-org.proxy2.library.illinois.edu/doi/10.1073/pnas.2411144122).
- 1008 <sup>48</sup> EPA, U. Bioretention Design Handbook: Designing Holistic Bioretention for Per-  
1009 formance and Longevity. EPA EPA 841-B-23-002 | (2023). URL [https://www.epa.](https://www.epa.gov/system/files/documents/2023-11/bioretentiondesignhandbook_plainnov2023.pdf)  
1010 [gov/system/files/documents/2023-11/bioretentiondesignhandbook\\_plainnov2023.](https://www.epa.gov/system/files/documents/2023-11/bioretentiondesignhandbook_plainnov2023.pdf)  
1011 [pdf](https://www.epa.gov/system/files/documents/2023-11/bioretentiondesignhandbook_plainnov2023.pdf).
- 1012 <sup>49</sup> New York City Department of Environmental Protection. New York City Stormwa-  
1013 ter Manual. Appendix (2024). URL [https://www.nyc.gov/assets/dep/downloads/](https://www.nyc.gov/assets/dep/downloads/pdf/water/stormwater/unified-stormwater-rule/uswr_nyc_stormwater_manual.pdf)  
1014 [pdf/water/stormwater/unified-stormwater-rule/uswr\\_nyc\\_stormwater\\_manual.](https://www.nyc.gov/assets/dep/downloads/pdf/water/stormwater/unified-stormwater-rule/uswr_nyc_stormwater_manual.pdf)  
1015 [pdf](https://www.nyc.gov/assets/dep/downloads/pdf/water/stormwater/unified-stormwater-rule/uswr_nyc_stormwater_manual.pdf).
- 1016 <sup>50</sup> City of Chicago Department of Water Management. City of Chicago  
1017 Stormwater Management Ordinance Manual. Tech. Rep. (2024). URL  
1018 [https://www.chicago.gov/content/dam/city/depts/water/general/Engineering/](https://www.chicago.gov/content/dam/city/depts/water/general/Engineering/SewerConstStormReq/2024StormwaterManual.pdf)  
1019 [SewerConstStormReq/2024StormwaterManual.pdf](https://www.chicago.gov/content/dam/city/depts/water/general/Engineering/SewerConstStormReq/2024StormwaterManual.pdf).
- 1020 <sup>51</sup> Hinman, C. & Extension, W. S. U. Rain Garden Handbook for Western Wash-  
1021 ington. Tech. Rep., Department of Ecology, State of Washington (2013). URL  
1022 <https://apps.ecology.wa.gov/publications/documents/1310027.pdf>.

- 1023 <sup>52</sup> New York State Department of Environmental Conservation & Center for Water-  
1024 shed Protection. New York State Stormwater Management Design Manual.  
1025 Tech. Rep. (2015). URL [https://cms4files1.revize.com/tonawandany/Engineer/  
1026 Stormwater%20Management%20Design%20Manual%20-%20Jan%202015.pdf](https://cms4files1.revize.com/tonawandany/Engineer/Stormwater%20Management%20Design%20Manual%20-%20Jan%202015.pdf).
- 1027 <sup>53</sup> Chin, D. A. *Water-Resources Engineering* 4th edn (Pearson, 2020).
- 1028 <sup>54</sup> Li, X. C., Zhao, L., Qin, Y., Oleson, K. & Zhang, Y. Elevated urban energy risks  
1029 due to climate-driven biophysical feedbacks. *Nature Climate Change* **14**, 1056–1063  
1030 (2024). URL <https://www.nature.com/articles/s41558-024-02108-w>.
- 1031 <sup>55</sup> Fitria, R., Kim, D., Baik, J. & Choi, M. Impact of Biophysical Mechanisms  
1032 on Urban Heat Island Associated with Climate Variation and Urban Morphol-  
1033 ogy. *Scientific Reports* **9**, 19503 (2019). URL [https://www.nature.com/articles/  
1034 s41598-019-55847-8](https://www.nature.com/articles/s41598-019-55847-8).
- 1035 <sup>56</sup> Mohammad Harmay, N. S. & Choi, M. Effects of heat waves on urban warming  
1036 across different urban morphologies and climate zones. *Building and Environ-*  
1037 *ment* **209**, 108677 (2022). URL [https://www.sciencedirect.com/science/article/  
1038 pii/S0360132321010672](https://www.sciencedirect.com/science/article/pii/S0360132321010672).
- 1039 <sup>57</sup> Lipson, M. J. *et al.* Evaluation of 30 urban land surface models in the Urban-  
1040 PLUMBER project: Phase 1 results. *Quarterly Journal of the Royal Meteorological*  
1041 *Society* **150**, 126–169 (2024). URL [https://onlinelibrary.wiley.com/doi/abs/10.  
1042 1002/qj.4589](https://onlinelibrary.wiley.com/doi/abs/10.1002/qj.4589).
- 1043 <sup>58</sup> Karsisto, P. *et al.* Seasonal surface urban energy balance and wintertime stabil-  
1044 ity simulated using three land-surface models in the high-latitude city Helsinki.  
1045 *Quarterly Journal of the Royal Meteorological Society* **142**, 401–417 (2016). URL  
1046 <https://onlinelibrary.wiley.com/doi/abs/10.1002/qj.2659>.

- 1047 <sup>59</sup> Lin, S., Feng, J., Wang, J. & Hu, Y. Modeling the contribution of long-term  
1048 urbanization to temperature increase in three extensive urban agglomerations in  
1049 China. *Journal of Geophysical Research: Atmospheres* **121**, 1683–1697 (2016). URL  
1050 <https://onlinelibrary.wiley.com/doi/abs/10.1002/2015JD024227>.
- 1051 <sup>60</sup> Zhang, L., Ye, Z. & Shibata, S. Assessment of Rain Garden Effects for the Man-  
1052 agement of Urban Storm Runoff in Japan. *Sustainability* **12**, 9982 (2020). URL  
1053 <https://www.mdpi.com/2071-1050/12/23/9982>.
- 1054 <sup>61</sup> Zhou, Z. & Guo, Q. Drainage Alternatives for Rain Gardens on Subsoil of Low  
1055 Permeability: Balance among Ponding Time, Soil Moisture, and Runoff Reduction.  
1056 *Journal of Sustainable Water in the Built Environment* **8**, 05022002 (2022). URL  
1057 <https://ascelibrary.org/doi/10.1061/JSWBAY.0000988>.
- 1058 <sup>62</sup> Winston, R. J., Dorsey, J. D. & Hunt, W. F. Quantifying volume reduction  
1059 and peak flow mitigation for three bioretention cells in clay soils in northeast  
1060 Ohio. *Science of The Total Environment* **553**, 83–95 (2016). URL <https://www.sciencedirect.com/science/article/pii/S0048969716302911>.
- 1062 <sup>63</sup> National Aeronautics and Space Administration. NLDAS-2 Forcing Dataset  
1063 Information | LDAS. URL <https://ldas.gsfc.nasa.gov/nldas/v2/forcing>.
- 1064 <sup>64</sup> Scharenbroch, B. C., Morgenroth, J. & Maule, B. Tree Species Suitability to  
1065 Bioswales and Impact on the Urban Water Budget. *Journal of Environmental Qual-  
1066 ity* **45**, 199–206 (2016). URL <https://onlinelibrary.wiley.com/doi/abs/10.2134/jeq2015.01.0060>.
- 1068 <sup>65</sup> Read, L. K., Hogue, T. S., Edgley, R., Mika, K. & Gold, M. Evaluating the Impacts  
1069 of Stormwater Management on Streamflow Regimes in the Los Angeles River. *Jour-  
1070 nal of Water Resources Planning and Management* **145**, 05019016 (2019). URL

- 1071 <https://ascelibrary.org/doi/10.1061/%28ASCE%29WR.1943-5452.0001092>.
- 1072 <sup>66</sup> Li, H., Sharkey, L. J., Hunt, W. F. & Davis, A. P. Mitigation of Impervious  
1073 Surface Hydrology Using Bioretention in North Carolina and Maryland. *Journal*  
1074 *of Hydrologic Engineering* **14**, 407–415 (2009). URL [https://ascelibrary.org/doi/](https://ascelibrary.org/doi/10.1061/%28ASCE%291084-0699%282009%2914%3A4%28407%29)  
1075 [10.1061/%28ASCE%291084-0699%282009%2914%3A4%28407%29](https://ascelibrary.org/doi/10.1061/%28ASCE%291084-0699%282009%2914%3A4%28407%29).
- 1076 <sup>67</sup> Nocco, M. A., Rouse, S. E. & Balster, N. J. Vegetation type alters water and  
1077 nitrogen budgets in a controlled, replicated experiment on residential-sized rain  
1078 gardens planted with prairie, shrub, and turfgrass. *Urban Ecosystems* **19**, 1665–  
1079 1691 (2016). URL <https://doi.org/10.1007/s11252-016-0568-7>.
- 1080 <sup>68</sup> Selbig, W. R. & Balster, N. Evaluation of Turf-Grass and Prairie-Vegetated Rain  
1081 Gardens in a Clay and Sand Soil, Madison, Wisconsin, Water Years 2004-2008.  
1082 Scientific Investigations Report 2010-5077, U.S. Department of Interior & U.S.  
1083 Geological Survey (2010).
- 1084 <sup>69</sup> Hathaway, J. M., Brown, R. A., Fu, J. S. & Hunt, W. F. Bioretention function  
1085 under climate change scenarios in North Carolina, USA. *Journal of Hydrology*  
1086 **519**, 503–511 (2014). URL [https://www.sciencedirect.com/science/article/pii/](https://www.sciencedirect.com/science/article/pii/S002216941400554X)  
1087 [S002216941400554X](https://www.sciencedirect.com/science/article/pii/S002216941400554X).
- 1088 <sup>70</sup> Avellaneda, P. M., Jefferson, A. J., Grieser, J. M. & Bush, S. A. Simulation of  
1089 the cumulative hydrological response to green infrastructure. *Water Resources*  
1090 *Research* **53**, 3087–3101 (2017). URL [https://onlinelibrary.wiley.com/doi/abs/10.](https://onlinelibrary.wiley.com/doi/abs/10.1002/2016WR019836)  
1091 [1002/2016WR019836](https://onlinelibrary.wiley.com/doi/abs/10.1002/2016WR019836).
- 1092 <sup>71</sup> DelVecchio, T., Welker, A. & Wadzuk, B. M. Exploration of Volume Reduction  
1093 via Infiltration and Evapotranspiration for Different Soil Types in Rain Garden  
1094 Lysimeters. *Journal of Sustainable Water in the Built Environment* **6**, 04019008

- (2020). URL <https://ascelibrary.org/doi/10.1061/JSWBAY.0000894>.
- <sup>72</sup> Hess, A., Wadzuk, B. & Welker, A. Evapotranspiration in Rain Gardens Using Weighing Lysimeters. *Journal of Irrigation and Drainage Engineering* **143**, 04017004 (2017). URL <https://ascelibrary.org/doi/10.1061/%28ASCE%29IR.1943-4774.0001157>.
- <sup>73</sup> Hess, A., Wadzuk, B. & Welker, A. Predictive Evapotranspiration Equations in Rain Gardens. *Journal of Irrigation and Drainage Engineering* **145**, 04019010 (2019). URL <https://ascelibrary.org/doi/10.1061/%28ASCE%29IR.1943-4774.0001389>.
- <sup>74</sup> McGauley, M. W., Amur, A., Shakya, M. & Wadzuk, B. M. A Complete Water Balance of a Rain Garden. *Water Resources Research* **59**, e2023WR035155 (2023). URL <https://onlinelibrary.wiley.com/doi/abs/10.1029/2023WR035155>. E2023WR035155 2023WR035155.
- <sup>75</sup> Wadzuk, B. M., Hickman, J. M. & Traver, R. G. Understanding the Role of Evapotranspiration in Bioretention: Mesocosm Study. *Journal of Sustainable Water in the Built Environment* **1**, 04014002 (2015). URL <https://ascelibrary.org/doi/10.1061/JSWBAY.0000794>.
- <sup>76</sup> Welker, A., Baghalian, S., Farnsworth, M. & Traver, R. Quantitative Analysis of the Water Budget of a Rain Garden in Pennsylvania. *Journal of Sustainable Water in the Built Environment* **10**, 04023013 (2024). URL <https://ascelibrary.org/doi/10.1061/JSWBAY.SWENG-545>.
- <sup>77</sup> University, V. Stormwater | Villanova University (2025). URL <https://www1.villanova.edu/villanova/sustainability/CampusSustainabilityBuildingsGroundsStormwaterDiningRecycling/>

1119 [StormwaterSustainability.html](#).

1120 <sup>78</sup> Villanova College of Engineering. Villanova Center for Resilient Water Sys-  
1121 tems | Villanova University (2025). URL [https://www1.villanova.edu/university/](https://www1.villanova.edu/university/engineering/faculty-research/Resilient-Water-Systems.html)  
1122 [engineering/faculty-research/Resilient-Water-Systems.html](https://www1.villanova.edu/university/engineering/faculty-research/Resilient-Water-Systems.html).

1123 <sup>79</sup> in *Chapter 17 - Terrestrial evapotranspiration* (eds Liang, S. & Wang, J.) *Advanced*  
1124 *Remote Sensing (Second Edition)* 649–684 (Academic Press, 2020). URL [https:](https://www.sciencedirect.com/science/article/pii/B9780128158265000179)  
1125 [//www.sciencedirect.com/science/article/pii/B9780128158265000179](https://www.sciencedirect.com/science/article/pii/B9780128158265000179).

1126 <sup>80</sup> Hart, D. Documentation | Derecho supercomputer (2023). URL [https://arc.ucar.](https://arc.ucar.edu/knowledge_base/74317833)  
1127 [edu/knowledge\\_base/74317833](https://arc.ucar.edu/knowledge_base/74317833).

1128 <sup>81</sup> Bureau, U. C. Cartographic Boundary Files - Shapefile. URL [https://www.census.](https://www.census.gov/geographies/mapping-files/time-series/geo/carto-boundary-file.html)  
1129 [gov/geographies/mapping-files/time-series/geo/carto-boundary-file.html](https://www.census.gov/geographies/mapping-files/time-series/geo/carto-boundary-file.html).

1130 <sup>82</sup> Beck, H. E. *et al.* High-resolution (1 km) Köppen-Geiger maps for 1901–2099  
1131 based on constrained CMIP6 projections. *Scientific Data* **10**, 724 (2023). URL  
1132 <https://www.nature.com/articles/s41597-023-02549-6>.

1133 <sup>83</sup> National Center for Atmospheric Research. CLM: Hydrologic Cycle (2020). URL  
1134 <https://www2.cesm.ucar.edu/models/clm/hydrologic.html>.

1135 <sup>84</sup> Guo, Y. & Gao, T. Analytical Equations for Estimating the Total Runoff Reduction  
1136 Efficiency of Infiltration Trenches. *Journal of Sustainable Water in the Built Envi-*  
1137 *ronment* **2**, 06016001 (2016). URL [https://ascelibrary.org/doi/10.1061/JSWBAY.](https://ascelibrary.org/doi/10.1061/JSWBAY.0000809)  
1138 [0000809](https://ascelibrary.org/doi/10.1061/JSWBAY.0000809).

1139 <sup>85</sup> Peng, S. *et al.* Climatic and biotic controls of evapotranspiration across grassland  
1140 ecosystems on the Tibetan Plateau. *Journal of Hydrology: Regional Stud-*  
1141 *ies* **61**, 102747 (2025). URL [https://www.sciencedirect.com/science/article/pii/](https://www.sciencedirect.com/science/article/pii/S2214581825005762)  
1142 [S2214581825005762](https://www.sciencedirect.com/science/article/pii/S2214581825005762).

- 1143 <sup>86</sup> Budyko, M. I. & Miller, D. H. Climate and life (1974).
- 1144 <sup>87</sup> National Oceanic and Atmospheric Administration. Climate Zones | National  
1145 Oceanic and Atmospheric Administration (2023). URL [https://www.noaa.gov/  
1146 jetstream/global/climate-zones](https://www.noaa.gov/jetstream/global/climate-zones).
- 1147 <sup>88</sup> Kang, W. *et al.* Characterization of Preferential Flow Occurrence Dur-  
1148 ing Freeze-Thaw Cycles. *Water Resources Research* **62**, e2025WR041926  
1149 (2026). URL <https://onlinelibrary.wiley.com/doi/abs/10.1029/2025WR041926>.  
1150 E2025WR041926 2025WR041926.
- 1151 <sup>89</sup> Wilby, R. L. & Keenan, R. Adapting to flood risk under climate change. *Progress  
1152 in Physical Geography* **36**, 348–378 (2012).
- 1153 <sup>90</sup> McGuire, M. P. *et al.* Retrofitting urban land through integrative, subsoils-based  
1154 planning of green stormwater infrastructure: A research framework. *Environmental  
1155 Research: Infrastructure and Sustainability* **1**, 035003 (2021). URL [https://dx.doi.  
1156 org/10.1088/2634-4505/ac27bd](https://dx.doi.org/10.1088/2634-4505/ac27bd).
- 1157 <sup>91</sup> UNEP-DHI Partnership Centre on Water and Environment *et al.* Green Infras-  
1158 tructure: Guide for Water Management. Tech. Rep., United Nations Environment  
1159 Programme (2014).
- 1160 <sup>92</sup> Schifman, L. A. *et al.* Situating Green Infrastructure in Context: A Framework for  
1161 Adaptive Socio-Hydrology in Cities. *Water Resources Research* **53**, 10139–10154  
1162 (2017).
- 1163 <sup>93</sup> McPhillips, L. E. & Matsler, A. M. Temporal Evolution of Green Stormwater  
1164 Infrastructure Strategies in Three US Cities. *Frontiers in Built Environment* **4**  
1165 (2018). URL [https://www.frontiersin.org/journals/built-environment/articles/10.  
1166 3389/fbuil.2018.00026/full](https://www.frontiersin.org/journals/built-environment/articles/10.3389/fbuil.2018.00026/full).

1167 <sup>94</sup> Xu, X. *et al.* Evaluation of variability among different precipitation products in the  
1168 Northern Great Plains. *Journal of Hydrology: Regional Studies* **24**, 100608 (2019).  
1169 URL <https://www.sciencedirect.com/science/article/pii/S2214581818303410>.

1170 <sup>95</sup> Singh, N. K., Emanuel, R. E., McGlynn, B. L. & Miniat, C. F. Soil Mois-  
1171 ture Responses to Rainfall: Implications for Runoff Generation. *Water Resources*  
1172 *Research* **57**, e2020WR028827 (2021). URL [https://onlinelibrary.wiley.com/doi/](https://onlinelibrary.wiley.com/doi/abs/10.1029/2020WR028827)  
1173 [abs/10.1029/2020WR028827](https://onlinelibrary.wiley.com/doi/abs/10.1029/2020WR028827). E2020WR028827 2020WR028827.

1174 <sup>96</sup> Herrmann, D. L., Schifman, L. A. & Shuster, W. D. Widespread loss of inter-  
1175 mediate soil horizons in urban landscapes. *Proceedings of the National Academy*  
1176 *of Sciences* **115**, 6751–6755 (2018). URL [https://www.pnas.org/doi/abs/10.1073/](https://www.pnas.org/doi/abs/10.1073/pnas.1800305115)  
1177 [pnas.1800305115](https://www.pnas.org/doi/abs/10.1073/pnas.1800305115).

1178 <sup>97</sup> Vodyanitskii, Yu. N. Organic matter of urban soils: A review. *Eurasian Soil Science*  
1179 **48**, 802–811 (2015). URL <https://doi.org/10.1134/S1064229315080116>.

1180 <sup>98</sup> Bergtold, J., Sailus, M. & Jackson, T. *Conservation Tillage Systems in the*  
1181 *Southeast: Production, Profitability and Stewardship* (2020).

1182 <sup>99</sup> Agency, M. P. C. Soil physical properties and processes | Minnesota Stormwater  
1183 Manual (2026). URL [https://stormwater.pca.state.mn.us/soil\\_physical\\_properties\\_](https://stormwater.pca.state.mn.us/soil_physical_properties_and_processes)  
1184 [and\\_processes](https://stormwater.pca.state.mn.us/soil_physical_properties_and_processes).

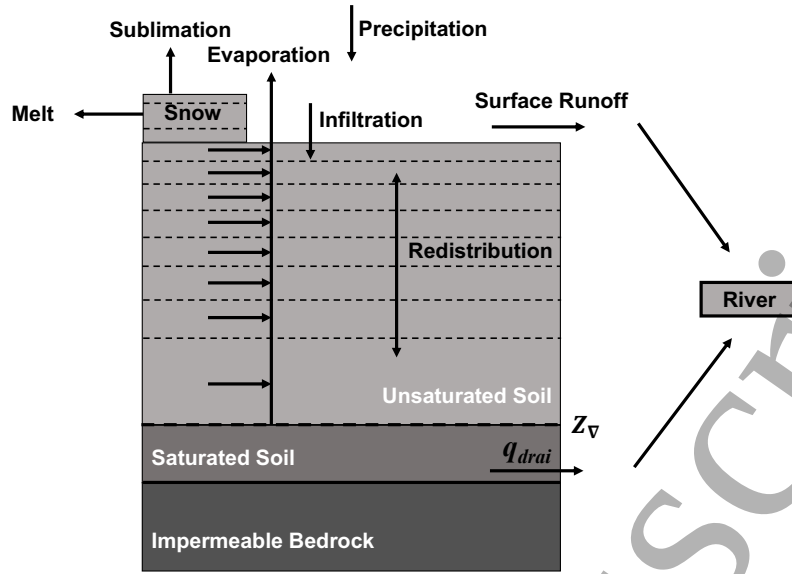
1185 <sup>100</sup> Lewellyn, C., Lyons, C. E., Traver, R. G. & Wadzuk, B. M. Evaluation of Seasonal  
1186 and Large Storm Runoff Volume Capture of an Infiltration Green Infrastructure  
1187 System. *Journal of Hydrologic Engineering* **21**, 04015047 (2016). URL [https:](https://ascelibrary.org/doi/10.1061/%28ASCE%29HE.1943-5584.0001257)  
1188 [/ascelibrary.org/doi/10.1061/%28ASCE%29HE.1943-5584.0001257](https://ascelibrary.org/doi/10.1061/%28ASCE%29HE.1943-5584.0001257).

1189 <sup>101</sup> Pineau, B. *et al.* Performance of green infrastructure for storm water treatment  
1190 in cold climate (Canada). *Journal of Environmental Engineering and Science* **16**,

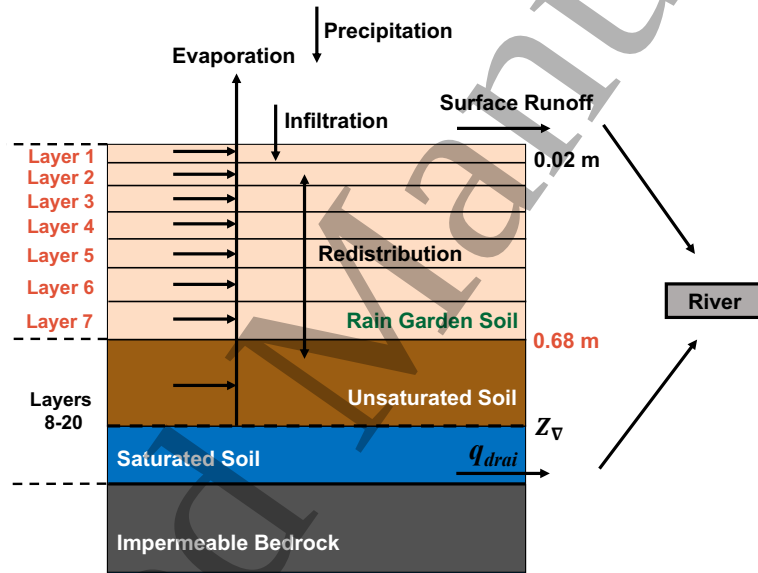
- 185–194 (2021). URL <https://doi.org/10.1680/jenes.20.00041>.
- <sup>102</sup> Sytsma, A., Crompton, O., Panos, C., Thompson, S. & Mathias Kondolf, G. Quantifying the Uncertainty Created by Non-Transferable Model Calibrations Across Climate and Land Cover Scenarios: A Case Study With SWMM. *Water Resources Research* **58**, e2021WR031603 (2022). URL <https://onlinelibrary.wiley.com/doi/abs/10.1029/2021WR031603>. E2021WR031603 2021WR031603.
- <sup>103</sup> Bethke, G. M., William, R. & Stillwell, A. S. Rain Garden Performance as a Function of Native Soil Parameters. *Journal of Sustainable Water in the Built Environment* **8**, 04021021 (2022). URL <https://ascelibrary.org/doi/10.1061/JSWBAY.0000967>.
- <sup>104</sup> Herrmann, D. L., Schiffman, L. A. & Shuster, W. D. Urbanization drives convergence in soil profile texture and carbon content. *Environmental Research Letters* **15**, 114001 (2020). URL <https://doi.org/10.1088/1748-9326/abbb00>.
- <sup>105</sup> Shuster, W. *et al.* K in an Urban World: New Contexts for Hydraulic Conductivity. *JAWRA Journal of the American Water Resources Association* **57**, 493–504 (2021). URL <https://onlinelibrary.wiley.com/doi/abs/10.1111/1752-1688.12918>.
- <sup>106</sup> Meenar, M., Heckert, M. & Adlakha, D. “Green Enough Ain’t Good Enough:” Public Perceptions and Emotions Related to Green Infrastructure in Environmental Justice Communities. *International Journal of Environmental Research and Public Health* **19**, 1448 (2022). URL <https://www.mdpi.com/1660-4601/19/3/1448>.
- <sup>107</sup> Whitten, M. in *Engaging Resilience: Integrating Sociocultural Dimensions into Green Infrastructure Planning* (eds Gomes Sant’Anna, C., Mell, I. & Schenk, L. B. M.) *Planning with Landscape: Green Infrastructure to Build Climate-Adapted Cities* 15–33 (Springer International Publishing, Cham, 2023). URL [https://doi.org/10.1007/978-3-031-25000-0\\_2](https://doi.org/10.1007/978-3-031-25000-0_2).

1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
21  
22  
23  
24  
25  
26  
27  
28  
29  
30  
31  
32  
33  
34  
35  
36  
37  
38  
39  
40  
41  
42  
43  
44  
45  
46  
47  
48  
49  
50  
51  
52  
53  
54  
55  
56  
57  
58  
59  
60

1215 [org/10.1007/978-3-031-18332-4\\_2](https://doi.org/10.1007/978-3-031-18332-4_2).

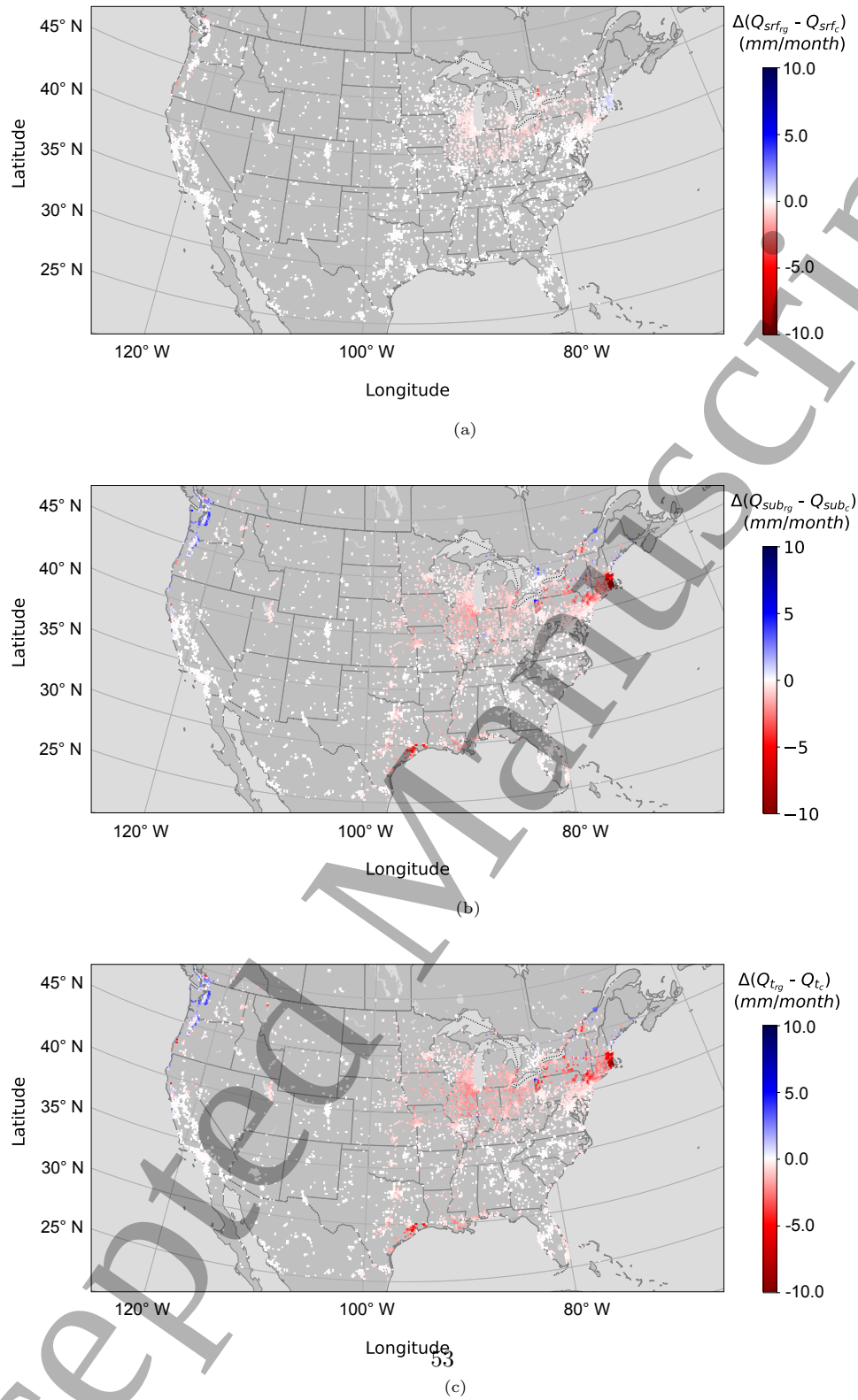


(a)

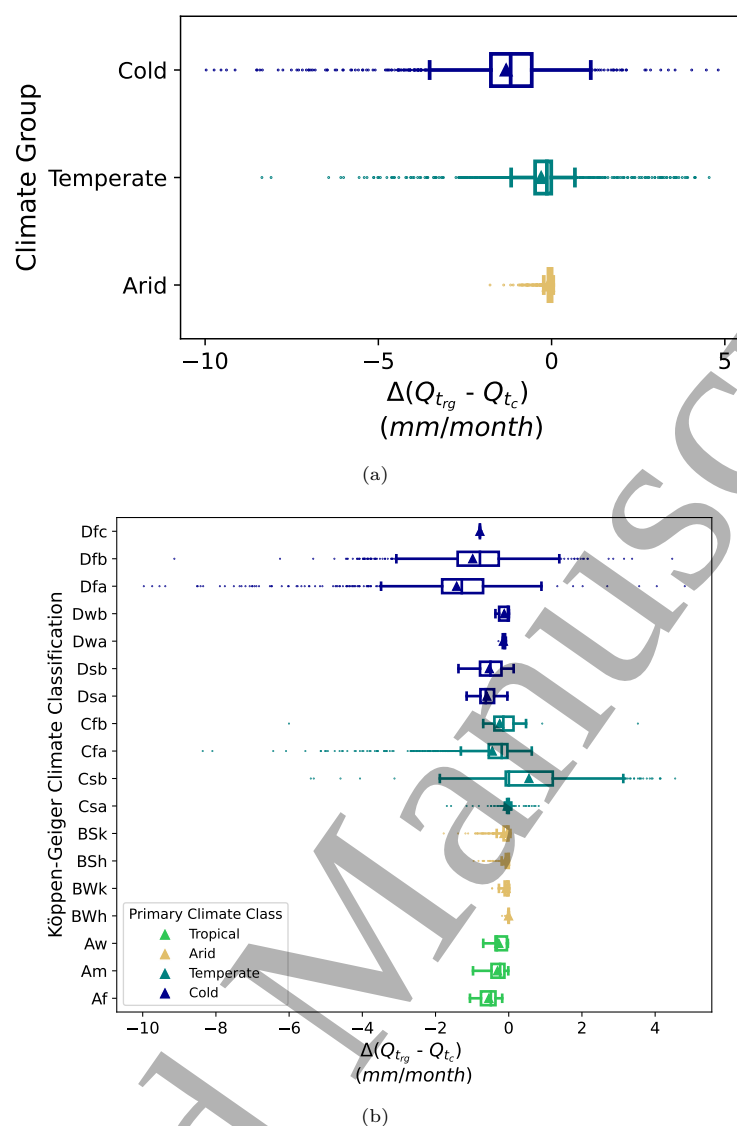


(b)

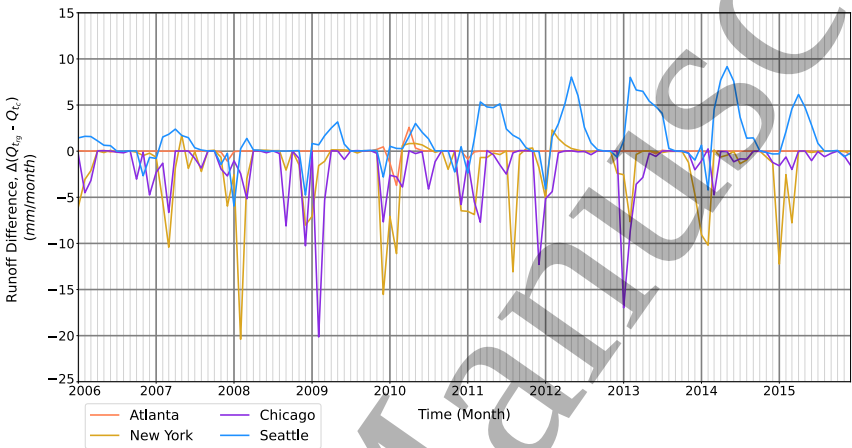
**Fig. 3:** Original hydrology for CLMU, recreated from<sup>44, 17</sup> (a) and modified hydrology for CLMU, reflecting rain garden implementation (b). Snow is still present in (b) but is excluded from this illustration for simplicity, and since we did not alter the existing snow scheme.  $q_{drain}$  refers to subsurface runoff. Both surface and subsurface runoff are routed to the river model after generation.



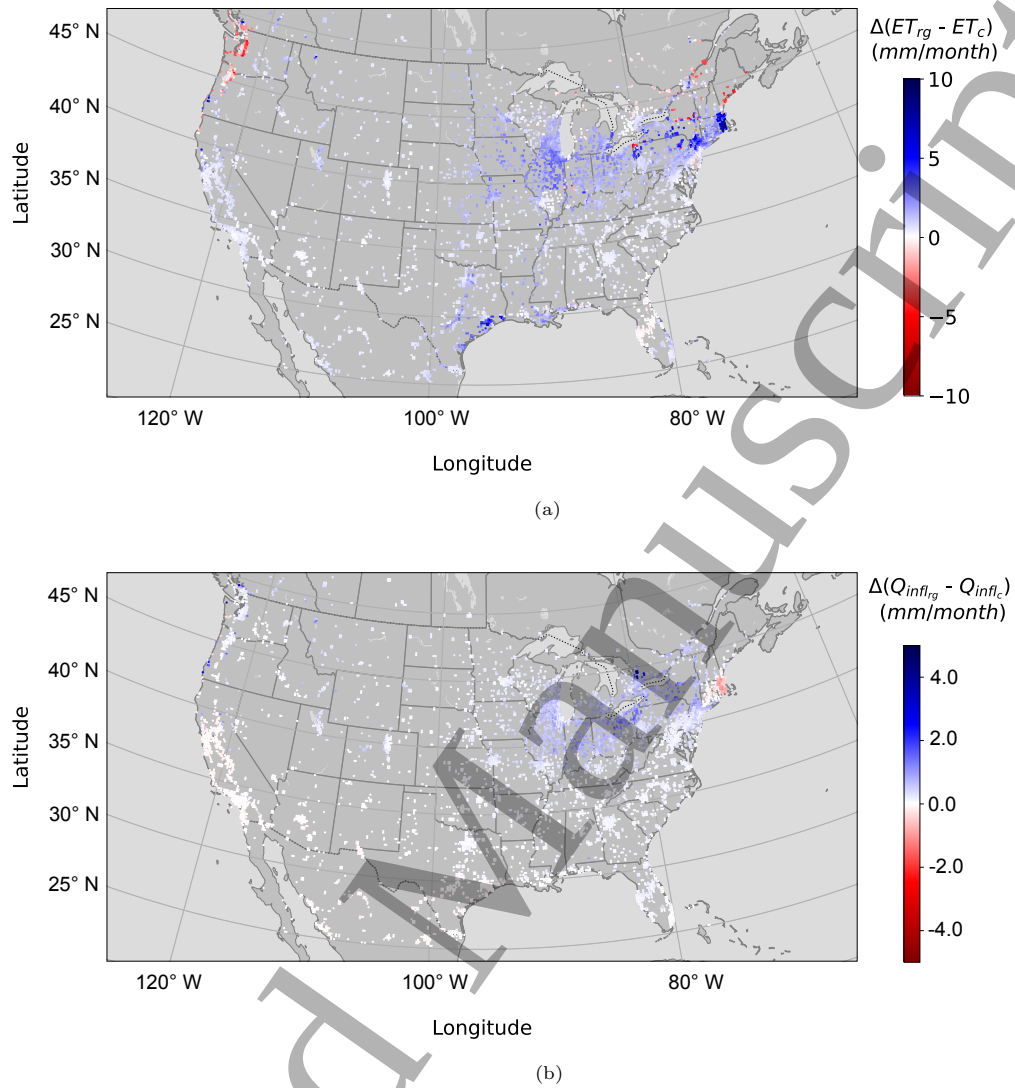
**Fig. 4:** Monthly average runoff differences observed across (a) urban surface runoff, (b) urban subsurface runoff, and (c) urban total runoff variables for the 10-year simulation period. Negative values indicate an average decrease in runoff in the rain garden simulation compared to the control simulation.



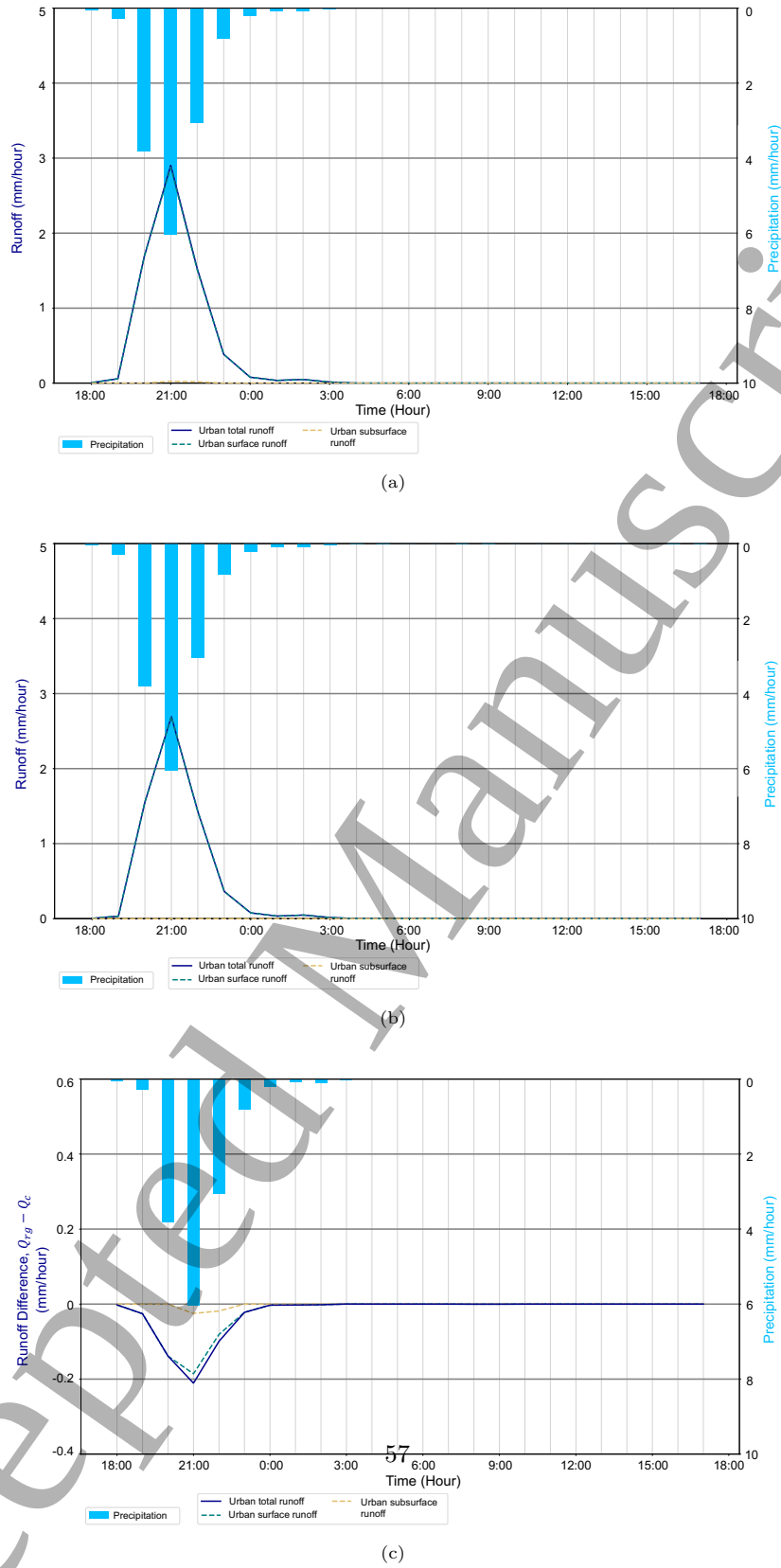
**Fig. 5:** Box plots illustrating the difference in mean urban runoff values (mm/month) for CONUS distributed according to (a) three climate groups (cold: continental + polar, temperate: temperate + tropic, and arid) and (b) more detailed Köppen-Geiger classification<sup>82</sup>. For (a), one-way analysis of variance (ANOVA) and Tukey pairwise testing reveals that runoff changes in each climate group are statistically significantly different from one another ( $p < 0.05$ ). For (b), the primary class is indicated by the color of the bar. For a breakdown of specific classes, see Supplemental Information Figure S10. The classes visualized represent the only classes present across CONUS urban areas.



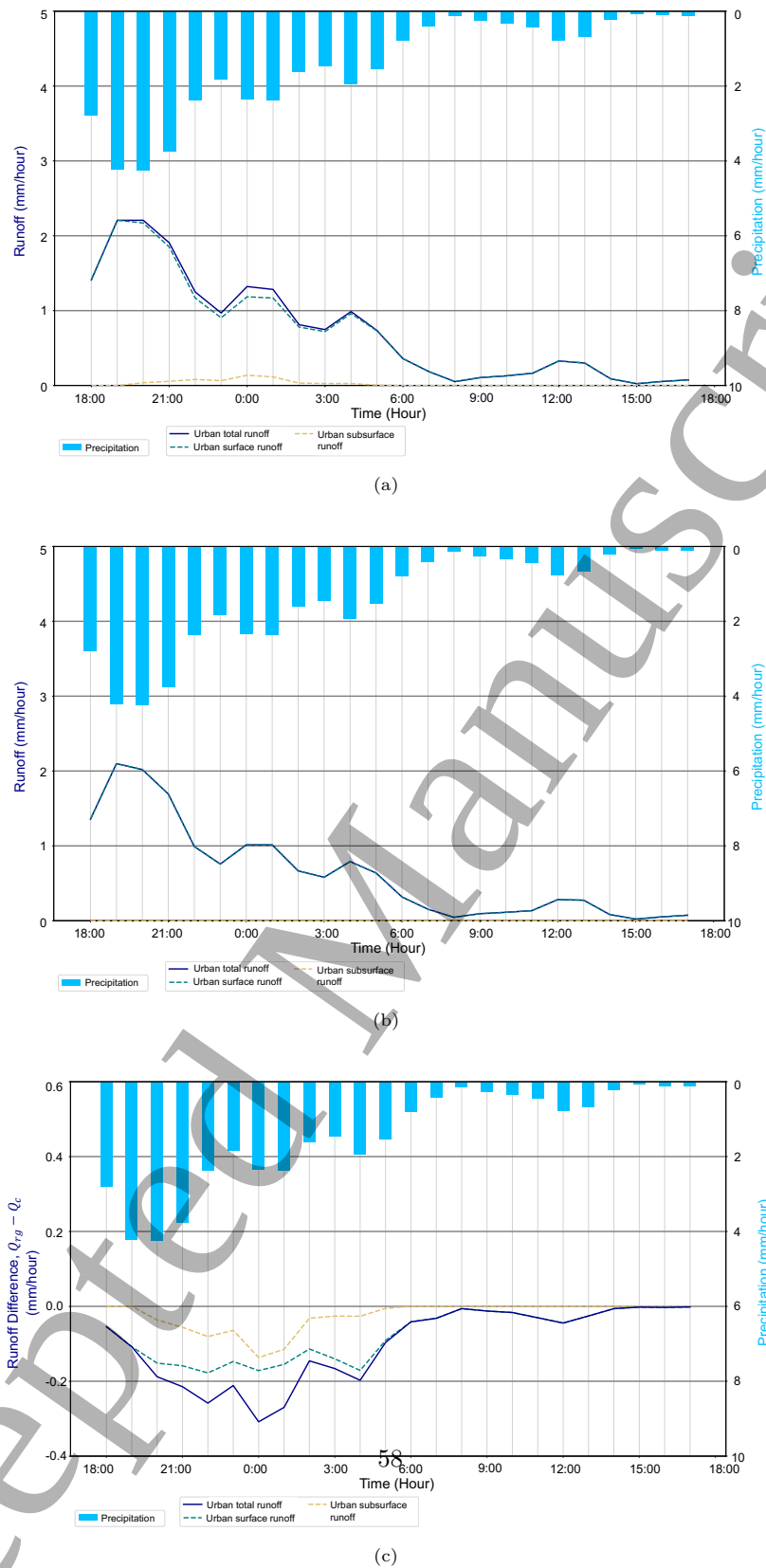
**Fig. 6:** Monthly average urban total runoff difference with rain gardens implemented for 2006–2015 for cities representing four U.S. regions: the Southeast (Atlanta), Midwest (Chicago), Northeast (New York), and West (Seattle).



**Fig. 7:** Monthly average urban (a) evapotranspiration difference and (b) infiltration difference observed for the 10-year simulation period. Positive values indicate an increase in either evapotranspiration or infiltration in the rain garden simulation compared to the control simulation.



**Fig. 8:** Comparison of urban runoff components generated from a Chicago spring storm event spanning April 19-20, 2011. Hourly precipitation is shown on the top axis. Urban total runoff,  $Q_t$ , represents the sum of urban surface runoff,  $Q_{surf}$ , and urban subsurface runoff,  $Q_{sub}$ . Subfigure (a) shows the control simulation, (b) shows the rain garden simulation, and (c) shows the difference in runoff generated between the two simulations ( $Q_{rg}-Q_c$ ), where negative differences in runoff indicate that there was less runoff in the rain garden simulation compared to the control simulation.



**Fig. 9:** Comparison of urban runoff components generated from a Chicago winter storm event spanning January 29-30, 2013. Urban total runoff,  $Q_t$ , represents the sum of urban surface runoff,  $Q_{surf}$ , and urban subsurface runoff,  $Q_{sub}$ . Subfigure (a) shows the control simulation, (b) shows the rain garden simulation, and (c) shows the difference in runoff generated between the two simulations ( $Q_{rg} - Q_c$ ), where negative differences in runoff indicate that there was less runoff in the rain garden simulation compared to the control simulation.